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The XianLiang Conjecture: Cosmic Redshift is Just the Light-Speed Expansion! $G(t)\,c(t)^2 = \boldsymbol{P}$, the Revised Einstein Field Equation $\boldsymbol{G}_{\mu\nu}$, and the Zero-Light-Speed Primordial Universe

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Abstract

Einstein's 1905 special relativity (SR) and 1915 general relativity (GR) demand only the strict Lorentz-frame invariance of light speed. They never require constant c across ten-billion-year cosmic timescales. This key distinction has been silently misread, for a century, by an untested cosmological assumption. In the 1983 SI definition, c was defined as 299792458 m/s, with a stated measurement uncertainty of ± 1.1 m/s, corresponding to a relative precision of approximately 3.67×10^{-9}. Critically, if the fractional daily growing rate of $c(t)$ is on the order of 10^{-13} day^{-1}, as predicted by the study's model, such a tiny variation lies far below the precision limit of the 1983 SI standard. Accordingly, it is entirely reasonable that this tiny drift has eluded all laboratory measurements over the full 43-year period from 1983 to 2026. Dirac's (1937) large numbers hypothesis entertained the possibility of time-varying gravitational parameter $G(t) \propto 1/t$ and light speed $c(t)$. Feynman (1960) and Davies (1995) both pointed out that the standard interpretation of cosmological redshift contains an unanswered causal gap in its physical mechanism. More recently, JWST (2023) and DESI (2025) data have revealed a series of theoretical conflicts with the current ΛCDM model, particularly at high redshift and in the nature of dark energy. In the spirit of doubt and inquiry pioneered by Dirac, Feynman, and Davies, the present study explores a new theoretical framework for cosmic evolution, built upon two core postulates: (1) the speed of light is time-dependent, while a cosmic coupling constant P remains invariant, satisfying $G(t)\,c(t)^2 = P$; and (2) the fine-structure constant α is eternally invariant. Under these two assumptions,

the time-growing speed of light $c(t)$ emerges as the sole driving force of cosmic evolution, and the gravitational constant G is no longer a fundamental constant but a derived physical quantity dependent on $c(t)$. The physical dimension of length, as defined by the SI meter, is re-examined within this framework. Vacuum permeability μ_0, vacuum permittivity ε_0, the Bohr radius a_{B} of hydrogen, the Schwarzschild radius r_s, the reduced Planck constant $\hbar$, the Planck mass m_{p}, the Planck length ℓ_{p}, the Planck density ρ_{p}, and the thermodynamic temperature T_{ion} in the Boltzmann ionization formula all evolve alongside $c(t)$. The time-dependent function $c(t)$ is inferred to scale synchronously with both the cosmic scale factor $a(t)$ and the redshift factor $1+z$ defined in the standard ΛCDM model. At the CMB decoupling epoch corresponding to $1+z=1100$, the speed of light $c(t)$ is $1/1100$ of the present speed c_0, the Bohr radius a_{B} is $1/1100$ of its current value, and the ionization temperature T_{ion} is only 82.64 K, a result markedly different from the conventional high-temperature assumption of the standard ΛCDM model. Within this growing-$c(t)$ model, the cosmic formation timescale was extended to $[-59.1, -45.2]$ Gyr, far exceeding the -13.8 Gyr assumed in the standard model, implying a gently evolving cosmos rather than a violent Big Bang scenario. The interpretation of the Einstein curvature field equation $G_{\mu\nu}$ is revisited, revealing an extremely high spacetime curvature in the early galaxy formation era ($1+z=25$) and the CMB epoch ($1+z=1100$). The intrinsic meaning of the cosmic coupling constant P is explored, with the relation $P = \dfrac{c(t)^6 G_{\mu\nu}}{8\pi T_{\mu\nu}} \approx \dfrac{c(t)^4 G_{\mu\nu}}{8\pi M_{\mu\nu}}$. Here, P characterizes the fundamental coupling between the cosmic mass-energy distribution, spacetime curvature and time-varying light speed $c(t)$. On macroscopic scales, the dynamic mass distribution governs the evolution of both spacetime curvature and light propagation velocity. The evolution of the speed of light encodes the transition of the universe from an early utterly homogeneous post-hadronization plasma state toward local clustering of matter and energy in spacetime. Its present fractional daily variation rate is calculated to be $[0.86, 1.89] \times 10^{-13}$ day^{-1}, and the feasibility of testing this variation is explored through both the High Energy Photon Source (HEPS) and the proposed Mission-M research initiative. Outstanding problems such as the DESI DR2 dynamical dark energy preference, the cosmic photon budget crisis, the early galaxy stellar mass tension, the Population III star timescale conflict, the early solar lithium abundance puzzle, and the late emergence of advanced intelligent life may be re-examined within this growing-$c(t)$ model. We postulate the ultimate cosmic origin as the *ZLZT* (Zero-Light-Speed and Zero-Temperature) primordial state, where both the speed of light and temperature approach zero. Meanwhile, the *LEM-SCDF* (Light-Electricity-Magnetism Synchronized Cosmic Dark Field) is defined to realize the global synchronization of electromagnetic parameters. More importantly, the revised Einstein field equation, which abandons the constant G and is equipped with dynamic $c(t)$, offers a groundbreaking perspective and feasible approach for exploring the long-sought unified field theory.

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References

1. Classical Basic Equations

This thesis is built upon several fundamental equations of physics, which lay the foundation for subsequent theoretical derivation. For clarity and completeness, these standard equations are briefly listed below with concise annotations, forming the baseline for our modified growing-$c(t)$ framework.

The speed of light in vacuum is related to the vacuum permeability and permittivity, derived from Maxwell's equations:

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \qquad \Rightarrow \qquad \mu_0 \varepsilon_0 c^2 = 1 \tag{1.1}$$

The fine-structure constant, which is a dimensionless fundamental constant, describes the electromagnetic coupling strength and constitutes the most elementary intrinsic quantity underlying the mass of all microscopic matter:

$$\alpha = \frac{e^2}{4\pi \varepsilon_0 \hbar c} \tag{1.2}$$

Planck mass, the fundamental unit of mass in natural units:

$$m_{\mathrm{P}} = \sqrt{\frac{\hbar c}{G}} \tag{1.3}$$

Planck length, the fundamental unit of length in natural units:

$$\ell_{\mathrm{P}} = \sqrt{\frac{\hbar G}{c^3}} \tag{1.4}$$

Planck density and Planck force, the characteristic density and force scales in quantum gravity, expressed in proportional form:

$$\rho_{\mathrm{P}} \propto \frac{m_{\mathrm{P}}}{{\ell_{\mathrm{P}}}^3} = \frac{c^5}{\hbar G^2}, \qquad F_{\mathrm{P}} = \frac{c^4}{G} \tag{1.5}$$

A phenomenological relation connecting microscopic Planck quantities and the macroscopic constant Gc^2. This formula is derived solely via dimensional analysis and lacks any physically meaningful underlying interpretation.

$$\frac{\hbar^4}{m_{\mathrm{P}}^5 \ell_{\mathrm{P}}^3} = \frac{\hbar^4 \rho_{\mathrm{P}}}{m_{\mathrm{P}}^6} = G c^2 \tag{1.6}$$

Bohr radius of the hydrogen atom, the most probable radius of the electron orbit in the ground-state hydrogen atom, is as follows:

$$a_{\mathrm{B}} = \frac{4\pi\varepsilon_0 \hbar^2}{m_e e^2} \tag{1.7}$$

Boltzmann ionization criterion for hydrogen atoms, expressed in two equivalent forms:

$$k_{\mathrm{B}} T_{\mathrm{ion}} > \frac{e^2}{4\pi\varepsilon_0 a_{\mathrm{B}}} = \alpha^2 m_e c^2 \tag{1.8}$$

Einstein curvature field equations of general relativity without cosmological constant:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \tag{1.9}$$

Einstein curvature field equations including the cosmological constant, describe the space-time curvature with dark energy contributions:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \tag{1.10}$$

The gravitational frequency shift in Schwarzschild spacetime, describe how the photon frequency changes with gravitational potential, and the Schwarzschild radius characterizes the scale of this effect:

$$\frac{\nu}{\nu_0} = \sqrt{1 - \frac{2GM}{c^2 r}}, \qquad r_s = \frac{2GM}{c^2} \tag{1.11}$$

Newton's law of universal gravitation, the classical description of gravitational force:

$$F_g = G\,\frac{m_1 m_2}{r^2} \tag{1.12}$$

Coulomb's law in vacuum for the electrostatic interaction between two point charges, where k_e denotes Coulomb's constant and ε_0 is the vacuum permittivity:

$$F_e = k_e \frac{q_1 q_2}{r^2}, \qquad k_e = \frac{1}{4\pi\varepsilon_0} \tag{1.13}$$

Friedmann equations derived from Einstein's field equations, describe the expansion of the universe. In this study, we define $\rho_{\text{F}}(t)$ as the mean cosmic density to distinguish it from the other density scenarios. The first equation defines the Hubble parameter $H(t) = \dot{a}(t)/a(t)$:

$$H^2(t) = \left(\frac{\dot{a}(t)}{a(t)}\right)^2 = \frac{8\pi G}{3}\rho_{\text{F}}(t) - \frac{kc^2}{a(t)^2} + \frac{\Lambda c^2}{3} \tag{1.14}$$

The second Friedmann equation, which governs the acceleration of cosmic expansion, include contributions from matter, radiation, and dark energy:

$$\frac{\ddot{a}(t)}{a(t)} = -\frac{4\pi G}{3}\left(\rho_{\text{F}}(t) + \frac{3p(t)}{c^2}\right) + \frac{\Lambda c^2}{3} \tag{1.15}$$

Two large-number coincidences were noted by Dirac. The first is the ratio of electric F_{em} to gravitational force F_{g} between an electron and a proton, the second is the ratio of the cosmic age T_0 to the light-crossing time of a proton.

$$N_{\text{Dirac,1}} = \frac{F_{\text{em}}}{F_{\text{g}}} = \frac{e^2}{4\pi\, G\, \varepsilon_0\, m_e\, m_p} \approx 2.27 \times 10^{39} \tag{1.16}$$

$$N_{\text{Dirac,2}} = \frac{T_0}{d_p/c} = \frac{c\,T_0}{d_p} \approx 7.67 \times 10^{40} \tag{1.17}$$

2. Fundamental Postulates

This study is built upon two core complementary postulates that define the theoretical framework for a time-evolving cosmos.

Macroscopic Postulate There exists a universal cosmic coupling constant $\boldsymbol{P}$ that remains invariant throughout the entire history of the universe. The speed of light $c(t)$ is not a constant but a time-dependent function of cosmic evolution, satisfying the fundamental conservation law:

$$G(t)\,c(t)^2 = \boldsymbol{P} \tag{2.1}$$

where $G(t)$ is the time-dependent gravitational coefficient. Under this postulate, $c(t)$ becomes the primary driver of cosmic evolution, and $G(t)$ is no longer a fundamental constant but a derived physical quantity.

Note: *Speed of light $c(t)$ is a global cosmic parameter. At any given time t, all photons in vacuum share the same speed $c(t)$, independent of the speed of light at the moment of their emission. This common speed is governed by an as-yet-unexplored cosmic dark field, which will be identified as the LEM-SCDF field in the subsequent sections.*

Microscopic Postulate The fine-structure constant $\boldsymbol{\alpha}$ is universally invariant throughout the post-CMB era, remaining constant from the epoch of neutral atom formation to the present day. In this framework, the defining expression is:

$$\boldsymbol{\alpha} = \frac{e^2}{4\pi\varepsilon_0(t)\,\hbar(t)\,c(t)} \tag{2.2}$$

remains invariant, even as $\varepsilon_0(t)$, $\hbar(t)$, and $c(t)$ themselves evolve with the cosmic time. This invariance is undefined in the primordial plasma phase before CMB decoupling era.

3. Symbol Definitions and Conventions

This section defines and standardizes the key symbols, subscript rules, and reference epochs used throughout the subsequent discussions, ensuring consistent notation and a clear physical interpretation.

Cosmic Reference Epochs (subscript z) Benchmark cosmic epochs are defined via redshift; subscript $*$ denotes the primordial epoch prior to the CMB epoch:

- $\boldsymbol{z_*}$: QGP confinement threshold epoch (Hadronization Epoch)
- $\boldsymbol{z_0}$: $1 + z = 1100$, CMB decoupling epoch
- $\boldsymbol{z_1}$: $1 + z = 25$, late reionization and early galaxy formation epoch
- $\boldsymbol{z_2}$: $1 + z = 3.7$, peak epoch of star formation and black hole activity, Cosmic Noon
- $\boldsymbol{z_3}$: $1 + z = 1.43$, Solar System formation epoch

Normalized Light Speed and Integrated Quantity Two core quantities were defined to characterize the time-varying cosmic light speed and its cumulative propagation effect:

- $\boldsymbol{C(t)}$: Dimensionless cosmic speed of light, defined as the ratio of the evolving light speed $c(t)$ to its present-day value c_0:

$$C(t) = \frac{c(t)}{c_0}$$

- $\boldsymbol{D(t_1, t_2)}$: Integrated effective light-travel time, obtained by performing a path integral of $C(t)$ over an arbitrary cosmic time interval $[t_1, t_2]$:

$$D(t_1, t_2) = \int_{t_1}^{t_2} C(t)\, dt$$

  $D(t_1, t_2)$ is measured in Gyr. When scaled by the present-day speed of light c_0, this time quantity corresponds to the standard light-travel distance in gigalight-years (Gly) adopted in conventional cosmology.

Light Speed Normalized light speed evaluated at each benchmark epoch:

- $\boldsymbol{C(t)|_{z_*}}$: Light speed at quark confinement threshold epoch (z_*)
- $\boldsymbol{C(t)|_{z_0}}$: Light speed at the CMB epoch (z_0)
- $\boldsymbol{C(t)|_{z_1}}$: Light speed at the early galaxy formation epoch (z_1)
- $\boldsymbol{C(t)|_{z_2}}$: Light speed at the Cosmic Noon epoch (z_2)
- $\boldsymbol{C(t)|_{z_3}}$: Light speed at the Solar System formation epoch (z_3)

Photon Intrinsic Frequencies Fixed intrinsic reference frequencies of photon, determined by atomic energy levels and the fine-structure constant:

- $\boldsymbol{\nu_{n0}} = 1.0 \times 10^{15}$ Hz: Primary intrinsic frequency

Medium Refractive Indices

- $\boldsymbol{N}_{\text{glass}} = 1.5$: Refractive index of standard optical glass (fixed)

Plasma Ionization Temperatures Critical Boltzmann ionization temperature for hydrogen atoms at each benchmark epoch:

- $\boldsymbol{T_{\text{ion}}|_{z_0}}$: Critical temperature at the CMB epoch (z_0)
- $\boldsymbol{T_{\text{ion}}|_{z_1}}$: Critical temperature at the early galaxy formation epoch (z_1)
- $\boldsymbol{T_{\text{ion}}|_{z_2}}$: Critical temperature at the Cosmic Noon epoch (z_2)
- $\boldsymbol{T_{\text{ion}}|_{z_3}}$: Critical temperature at the Solar System formation epoch (z_3)

Einstein Spacetime Curvature Einstein curvature tensor evaluated at each benchmark epoch:

- $\boldsymbol{G_{\mu\nu}|_{z_0}}$: Curvature at the CMB epoch (z_0)
- $\boldsymbol{G_{\mu\nu}|_{z_1}}$: Curvature at the early galaxy formation epoch (z_1)
- $\boldsymbol{G_{\mu\nu}|_{z_2}}$: Curvature at the Cosmic Noon epoch (z_2)
- $\boldsymbol{G_{\mu\nu}|_{z_3}}$: Curvature at the Solar System formation epoch (z_3)

Present-Day Cosmic Reference Values (Subscript 0) Fundamental physical constants and cosmic scales evaluated at the current cosmic epoch, serving as the universal reference baseline throughout this work:

- $\boldsymbol{c_0}$: Present-day speed of light in vacuum
- $\boldsymbol{G_0}$: Present-day gravitational constant
- $\boldsymbol{a_0}$: Present-day cosmic scale factor (Friedmann metric)

Three Definitions of Cosmic Time We introduce three distinct cosmic time variables to distinguish between the *variable speed-of-light cosmology* developed in this work and the standard *constant speed-of-light cosmology*:

- $\boldsymbol{t}$: Cosmic time in the *variable speed-of-light framework*. By convention, $t \leq 0$ everywhere, with the *present epoch* set to $t = 0$. More ancient cosmic epochs corresponded to *more negative t-values* of t. The unit of t is Gyr (gigayears).
- $\boldsymbol{t^\star}$: Cosmic time in the *standard constant speed-of-light cosmology*. This is the conventional cosmic time used in ΛCDM, also defined such that $t^\star \leq 0$ and the present epoch is $t^\star = 0$. The epoch of *Cosmic Noon* (the peak of cosmic star-formation activity) corresponds to $t^\star \approx -11.4\,\mathrm{Gyr}$ in standard cosmology.
- $\boldsymbol{t^\Delta}$: Forward cosmic time defined within the standard ΛCDM model. It takes the moment of the Big Bang as the origin $t^\Delta = 0$, so we always have $t^\Delta \geq 0$. The present epoch corresponded to $t^\Delta = 13.8\,\mathrm{Gyr}$.

  In particular, the epoch of *Cosmic Noon* is located at $t^\Delta = 2.4\,\mathrm{Gyr}$ in this forward time system. These two standard cosmological time coordinates satisfy:

$$t^\Delta + |t^\star| = 13.8\,\mathrm{Gyr}. \tag{3.1}$$

  Note: *Superscript $\star$ indicates quantities under standard constant-speed cosmology with the present as time zero. The superscript $\triangle$ denotes the forward ΛCDM time starting from Big Bang, which is commonly used for redshift analysis. This notation distinguishes our framework from conventional cosmology.*

4. The Overlooked Temporal Variability of Light Speed

The Axiomatic Trap For over a century, the constancy of the speed of light c has been elevated to the status of an unshakable dogma. While Einstein's theories achieved remarkable success in local regimes, they inadvertently imposed a cognitive confinement: researchers habitually project laboratory-scale measurements onto the entire cosmic timeline. Such axiomatic inertia leaves few scholars bold enough to explore the prospect of variable light speed.

Yet Einstein's 1905 special relativity (SR) and 1915 general relativity (GR) demand only the ***strict*** Lorentz-frame invariance of c — they ***never*** require constancy of c across ten-billion-year cosmic epochs. This subtle distinction has been silently misread for a century as a universal cosmological assumption. Since Rømer (1676) first pointed out that light travels at finite speed, human measurements of light speed c have spanned less than four centuries — a mere heartbeat to the cosmic giant.

In the 1983 SI revision, the speed of light was fixed at 299 792 458m/s, with a relative precision documented in the NBS report [1] at approximately 3.67×10^{-9}. Critically, if the fractional daily growth rate of $c(t)$ were on the order of 10^{-13} day^{-1}, as suggested by the subsequent models in Section 5 and Section 6, then no experiment would have been able to detect such a tiny drift over the 43 years since 1983, because it lies far below the 1983 standard's precision limit.

Observational Asymmetry: The Fundamental Blind Spot Modern cosmology rests on a conspicuous yet long neglected asymmetry. Ancient photons and their redshift phenomena are observable, but we possess ***zero*** direct evidence for the actual speed of light c and gravitational strength G in the early universe. The standard model reduces all consequences of cosmic expansion to a single observable phenomenon of frequency redshift, while blindly holding the speed of light c fixed permanently. This judgment lacks practical verification, merely serving as an unsubstantiated metaphysical presumption that has never undergone rigorous examination.

Topological Necessity: The 2D Geometric Imperative Here lies the fatal geometric flaw in the standard cosmological framework. Cosmic expansion is not a one-dimensional stretching of a line, but uniform scaling deformation of a curved two-dimensional spacetime manifold. Riemannian geometry and topology strictly dictate that the isotropic expansion of a metric surface must distribute its deformation across all inherent degrees of freedom. Light possesses two mutually conjugate kinematic properties: intrinsic frequency and propagation velocity. To assign the entire metric deformation to frequency alone is a mathematical anomaly. The standard assumption effectively amputates half degrees of freedom of spacetime, resulting in a physically incomplete and topologically inconsistent description.

The Symmetry-Restoring Postulate To repair this broken geometric symmetry, the most parsimonious and mathematically inevitable conclusion is:

$$\frac{\boldsymbol{\nu_0}}{\boldsymbol{\nu_e}} \propto \frac{\boldsymbol{a(t_e)}}{\boldsymbol{a_0}} \quad \Rightarrow \quad \frac{\boldsymbol{c(t)}}{\boldsymbol{c_0}} \propto \frac{\boldsymbol{a(t)}}{\boldsymbol{a_0}} \tag{4.1}$$

The speed of light must evolve synchronously with the cosmic scale factor. This is not merely a physical hypothesis; it is a requirement to restore the topological integrity of the expanding universe.

Note: *In the subsequent Section 10, by challenging the* ***causal gap*** *within standard cosmological redshift theory, we further argue that cosmic redshift is just the light-speed expansion.*

In the following three sections, we construct two different growing-$c(t)$ models: the *sigmoid*-like compact model in Section 5 and the `Planck18`-calibrated ultra-fine model in Section 6. Then, in Section 7, we seek to analyze the dramatically different physical pictures emerging under the growing-$c(t)$ assumption across various cosmic epochs.

5. Sigmoid-like Growing-$c(t)$ Model

This section introduces a physically motivated *sigmoid*-like model for the time-dependent speed of light $c(t)$, which seeks to describe the naturally gradual rise of light speed from the early universe to the present.

The Standard Sigmoid Function The sigmoid function is a smooth, monotonic, and bounded function widely applied in biology, botany, life sciences, and data science to describe fundamental growth, maturation, saturation, and transition phenomena. Its standard form is

$$\sigma(x) = \frac{1}{1 + e^{-x}} \ , \tag{5.1}$$

where $\sigma(x) \in (0, 1)$ for all real x. This function is symmetric about point $(0,\, 0.5)$, and increases steadily over the entire real domain. For clarity, Table 1 lists the key numerical values at representative points.

Table 1: Values of the standard sigmoid function $\sigma(x)$

$\sigma(x)$	0.047	0.119	0.269	0.378	**0.500**	0.622	0.731	0.881	0.953
x	-3.0	-2.0	-1.0	-0.5	0.0	$+0.5$	$+1.0$	$+2.0$	$+3.0$

The following is the standard mathematical curvature $\kappa(x)$ expression of the sigmoid $\sigma(x)$ function without detailed derivation:

$$\kappa(x) = \frac{|\ddot{\sigma}(x)|}{\big[1 + (\dot{\sigma}(x))^2\big]^{3/2}} = \frac{\big|\sigma(x)\big[1 - \sigma(x)\big]\big[1 - 2\sigma(x)\big]\big|}{\big\{1 + \big[\sigma(x)\big(1 - \sigma(x)\big)\big]^2\big\}^{3/2}} \ , \tag{5.2}$$

The curvature function $\kappa(x)$ has three key points: two symmetric maxima and one minimum. The left maximum occurs at $x = -\ln(2 + \sqrt{3}) \approx -1.317$, and the right maximum at $x = \ln(2 + \sqrt{3}) \approx +1.317$. The curvature equals zero at $x = 0$, which is the inflection point where the curve changes concavity. As $x \to \pm\infty$, the curvature asymptotically approaches zero.

Notably, if the cosmic redshift function $z(t)$ takes the exponential form $z(t) \propto e^{-t}$, then the quantity $1/\big(1 + z(t)\big)$ closely resembles the standard sigmoid function $\sigma(x) = 1/\big(1 + e^{-x}\big)$. If this correspondence holds, it would build a natural bridge between sigmoid formalism and cosmic redshift evolution, laying a solid foundation for the analysis described in the following subsection.

Redshift Curve $z(t)$: Morphology and Calibration We obtain the standard ΛCDM redshift–time relation using the Python library `astropy.cosmology` [2], specifically the `Planck18` cosmological implementation [3], which provides numerically consistent redshift values as a function of forward cosmic time t^{Δ} elapsed since the Big Bang.

A routinely calibrated cornerstone in observational studies is the so-called *Cosmic Noon* epoch, corresponding to $t^\Delta = 2.4\,\mathrm{Gyr}$ with a matched redshift $z = 2.7$. We adopt this reference to build the simplified exponential model:

$$z(t^\Delta) = \underbrace{A^\Delta\, e^{-k^\Delta\left(t^\Delta - \beta^\Delta\right)}}_{\text{Exp. decay}} \tag{5.3}$$

parameters are set as

$$A^\Delta = 2.7, \quad \beta^\Delta = 2.4\,\mathrm{Gyr}, \quad k^\Delta = 1.0\ , \tag{5.4}$$

such that the argument of the exponential equals zero exactly at the Cosmic Noon epoch:

$$t^\Delta - \beta^\Delta = 0 \quad \Rightarrow \quad e^0 = 1 \quad \Rightarrow \quad z(t^\Delta) = A^\Delta\ . \tag{5.5}$$

While the above simplified model captures the general trend of cosmic redshift, we construct an improved composite model to achieve a better fit to the ΛCDM `Planck18` numerical data. We extend the basic formulation by incorporating two additional terms: a hyperbolic correction and a pseudo-elliptic adjustment. The full expression reads:

$$z(t^\Delta) = \underbrace{A^\Delta\, e^{-k^\Delta\left(t^\Delta - \beta^\Delta\right)}}_{\text{Exp. decay}} \cdot \underbrace{\left(\frac{t^\Delta}{\beta^\Delta}\right)^{k^\Delta - 1}}_{\text{Hyperbolic corr.}} \cdot \underbrace{\left(\frac{13.8 - t^\Delta}{13.8 - \beta^\Delta}\right)^{1 - k^\Delta}}_{\text{Pseudo-elliptic adj.}} \tag{5.6}$$

The hyperbolic correction slows the decay rate and brings the profile closer to the ΛCDM morphology. The two additional terms employ complementary exponents to avoid numerical singularities and zero-denominator issues. The pseudo-elliptic term enforces the present-day boundary condition $z(t^\Delta) = 0$ at $t^\Delta = 13.8\,\mathrm{Gyr}$, a feature unattainable for exponential model.

Despite its complex appearance, the improved model contains only one independent free parameter k^Δ. The set of calibration parameters adopted for global morphological tuning is given by

$$A^\Delta = 2.7, \quad \beta^\Delta = 2.4\,\mathrm{Gyr}, \quad k^\Delta = 0.16\ , \tag{5.7}$$

The figure below compares the redshift curves obtained from the two models.

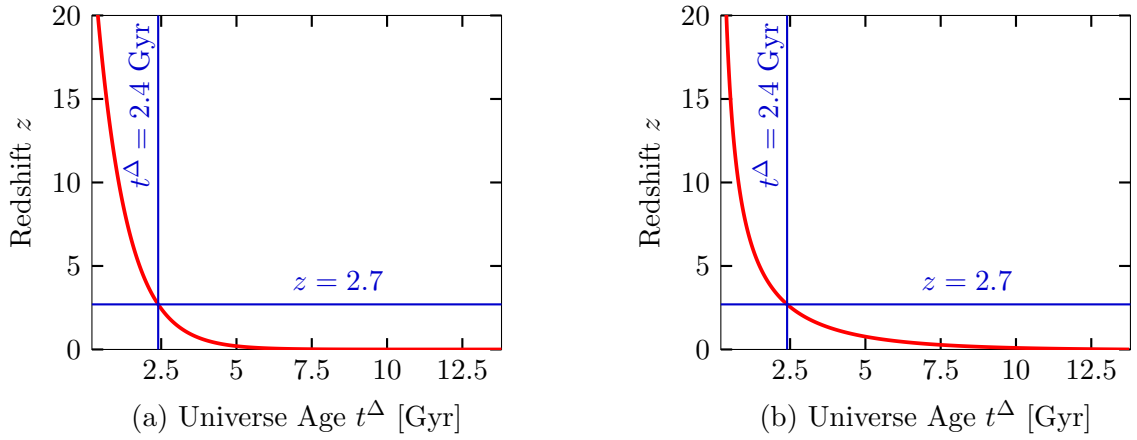


Figure 1: Redshift $z(t^\Delta)$ profiles using two models: (a) Simple exponential model Eq. (5.3); (b) improved composite model with exponential, hyperbolic, and pseudo-elliptic components Eq. (5.6). Both models are anchored at the reference point $t^\Delta = 2.4$ Gyr and $z = 2.7$.

Both the simple exponential model Eq. (5.3) and the improved composite model Eq. (5.6) well reproduce the exponential decay morphology of the observed redshift curve. Meanwhile, the improved model achieves a superior overall fit to ΛCDM `Planck18` data across all cosmic epochs.

Nevertheless, we conventionally adopt the simple exponential ansatz $z(t) \propto e^{-t}$ to proceed with our study on time-varying light speed. The relation between $c(t)$, scale factor $a(t)$ and redshift $z(t)$ is given below:

$$\frac{\boldsymbol{c(t)}}{\boldsymbol{c_0}} = \frac{\boldsymbol{a(t)}}{\boldsymbol{a_0}} = \frac{\mathbf{1}}{\mathbf{1} + \boldsymbol{z(t)}} \ , \tag{5.8}$$

We adopt this simple formulation to construct $c(t)$ model for the following reasons:

- It has a compact mathematical formulation and captures the dominant evolutionary trend of cosmic redshift.
- The cosmic time and expansion history $z(t^\Delta)$ derived from standard ΛCDM `Planck18` data are theoretically inferred under the assumption of a constant light speed.
- To date, no direct observational evidence reveals how the speed of light evolved throughout cosmic history if c is indeed time-dependent.
- Adopting a time-varying light speed, our theoretical framework is fundamentally distinct from standard cosmology with a constant speed of light.
- The smooth sigmoid function is a reasonable extrapolation from the redshift $z(t)$ data; its favorable mathematical properties will also facilitate our toy model construction and theoretical analysis.

The Growing $c(t)$ Model Our model assumes that the speed of light continuously rises from the early universe to the present day, and will keep increasing slowly in the future. In this framework, we set the cosmic time of the present day to $t = 0$, while earlier cosmic times take larger negative values. This time convention differs from the variable t^{Δ} adopted in the preceding redshift model.

Accordingly, we describe the evolving $c(t)$ using a generalized sigmoid function, which features local symmetry about its inflection point:

$$\frac{c(t)}{c_0} = \frac{1}{1 + e^{-k(t+\beta)}} \ . \tag{5.9}$$

Here, k and β are two positive parameters to be determined subsequently. In the extremely early universe, as $t \to -\infty$, the exponential term $e^{-k(t+\beta)} \to +\infty$, which means the speed of light back then was far lower than its present value c_0.

To normalize the function for the present epoch ($t = 0$), where the condition $c(0)/c_0 = 1$ is enforced, a constant factor $(1+e^{-k\beta})$ is added to the numerator. This yields the final normalized formula:

$$\boldsymbol{C(t)} = \frac{c(t)}{c_0} = \frac{1 + e^{-k\beta}}{1 + e^{-k(t+\beta)}} \ . \tag{5.10}$$

Integral and Derivatives of $C(t)$ To evaluate the effective light travel distance across arbitrary cosmic epochs, we first present the definite integral of the normalized light speed function Eq. (5.10) over a general time interval $[t_1, t_2]$. This integral is essential for comparing our model with observational cosmic timescales.

Introduce a constant to simplify the normalized profile in Eq. (5.10):

$$J = 1 + e^{-k\beta} \ , \tag{5.11}$$

so that

$$\frac{c(t)}{c_0} = \frac{J}{1 + e^{-k(t+\beta)}} \ . \tag{5.12}$$

The **definite integral** over $[t_1, t_2]$ is given by:

$$D(t_1, t_2) = \int_{t_1}^{t_2} \frac{c(t)}{c_0} \, \mathrm{d}t = \frac{J}{k} \Big[k(t+\beta) + \ln\left(1 + e^{-k(t+\beta)}\right) \Big] \Bigg|_{t_1}^{t_2} . \tag{5.13}$$

We further derive the first and second order derivatives, which describe respectively the variation rate and acceleration of the evolving light speed.

The **first derivative** (rate of light speed variation) takes the form:

$$\dot{C}(t) = \frac{d}{dt}\left(\frac{c(t)}{c_0}\right) = \frac{k\,J\,e^{-k(t+\beta)}}{\left(1+e^{-k(t+\beta)}\right)^2}\ . \tag{5.14}$$

For the **second derivative** (acceleration of light speed variation), we have:

$$\ddot{C}(t) = \frac{d^2}{dt^2}\left(\frac{c(t)}{c_0}\right) = \frac{k^2\,J\,e^{-k(t+\beta)}\left(e^{-k(t+\beta)}-1\right)}{\left(1+e^{-k(t+\beta)}\right)^3}\ . \tag{5.15}$$

The inflection point denoted as t_{inf} :

$$t_{\text{inf}} = -\beta\ . \tag{5.16}$$

These expressions hold for all admissible positive k and β.

The Left Maximum Curvature Point of $C(t)$ Referring to the first and second derivatives of $C(t)$ defined in Eq. (5.14) and Eq. (5.15) , we now derive the curvature function for the normalized light speed profile.

Following the standard definition of curvature introduced in Eq. (5.2), the curvature $\kappa(t)$ is always non-negative because its numerator contains an absolute value. Therefore, a point where the first derivative of $\kappa(t)$ equals zero corresponds to a local maximum curvature.

Substituting the derivatives into the curvature formula yields the explicit expression for $\kappa(t)$:

$$\kappa(t) = \frac{\left|\ddot{C}(t)\right|}{\left[1+(\dot{C}(t))^2\right]^{3/2}} = \frac{\left|\dfrac{k^2 J e^{-k(t+\beta)}\left(e^{-k(t+\beta)}-1\right)}{\left(1+e^{-k(t+\beta)}\right)^3}\right|}{\left[1+\left(\dfrac{kJe^{-k(t+\beta)}}{\left(1+e^{-k(t+\beta)}\right)^2}\right)^2\right]^{3/2}}\ , \tag{5.17}$$

Instead of solving this complex expression directly, we resort to an approximate scheme. The parameters k and J slightly break the exact translational invariance of the sigmoid profile, but valid approximate invariance remains satisfied under the condition $kJ \approx 1$. The left maximum curvature point can then be approximated by

$$t_{\text{leftmax}} \approx -\beta - \frac{\ln(2+\sqrt{3})}{k}\ . \tag{5.18}$$

Parameter Solving via Cosmic Noon Anchoring Since the genuine evolutionary curve of cosmic light speed cannot be directly observed, we select the most physically meaningful cosmic epoch as the anchor point to solve the parameters $\boldsymbol{k}$ and $\boldsymbol{\beta}$ in function Eq. (5.10).

Cosmic Noon corresponds to the epoch of peak cosmic star formation at $t^\star \approx -11.4$ Gyr, marking the universe's evolutionary rhythm at its most vigorous growth stage, coinciding with the curvature maximum of the model's evolutionary curve, and serving as the most representative temporal anchor point.

Denoting the cosmic noon time by $\boldsymbol{t_w}$, we impose three independent physical constraints:

- The effective light-travel integral from Eq. (5.13) satisfies $D(t_w, 0) = 11.4\text{Gyr}$, equal to the integrated light propagation distance spanning cosmic noon to the present day.
- The normalized light speed at cosmic noon reads $C(t_w) = 1/3.7$, constrained by the measured redshift $z = 2.7$ together with the model definition in Eq. (5.10).
- Based on physical self-consistency, cosmic noon coincides with the left curvature maximum from Eq. (5.18), i.e., $t_w = t_{\text{leftmax}}$, where the acceleration change rate of $c(t)$ reaches its peak.

These three constraints form a closed algebraic system for the three unknown variables $\boldsymbol{k}$, $\boldsymbol{\beta}$ and $\boldsymbol{t_w}$, explicitly written as

$$\begin{cases} \dfrac{J}{k}\left[k(t+\beta) + \ln\left(1 + e^{-k(t+\beta)}\right)\right]\Big|_{t_w}^{0} &= 11.4 \\ \dfrac{1 + e^{-k\beta}}{1 + e^{-k(t_w+\beta)}} &= \dfrac{1}{3.7} \\ -\beta - \dfrac{\ln(2+\sqrt{3})}{k} &= t_w \end{cases} \tag{5.19}$$

The resulting parameters read

$$\boldsymbol{k} = 0.1442, \quad \boldsymbol{\beta} = 8.854, \quad \boldsymbol{t_w} = -17.98\,\text{Gyr}, \quad \boldsymbol{J} = 1.2789. \tag{5.20}$$

Up to this point, the compact growing-$c(t)$ model has been fully established. We now present its evolution curve in the following figure.

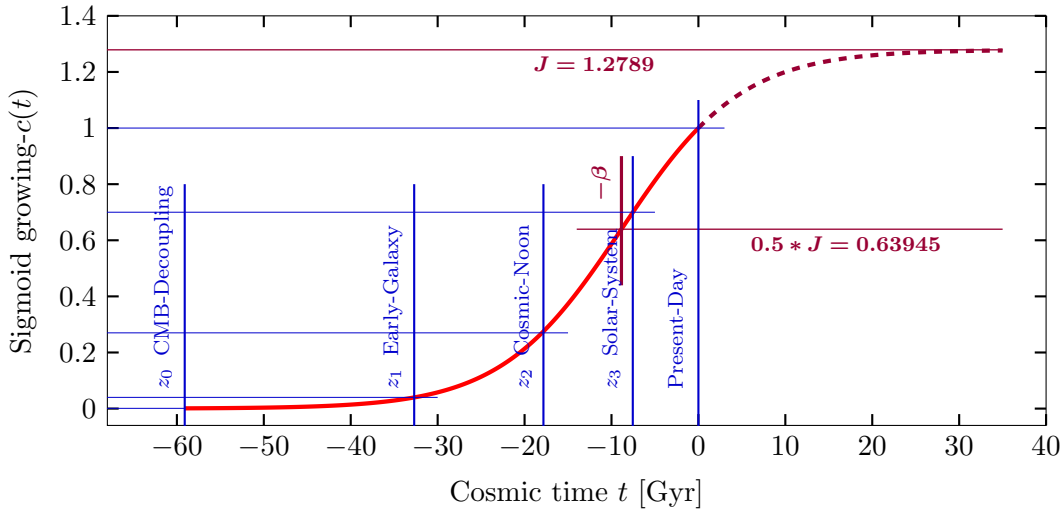


Figure 2: The C(t) curve of the compact sigmoid growing-$c(t)$ model with determined parameters $k = 0.1442$, $\beta = 8.854$, $t_w = -17.98\,\text{Gyr}$, and $J = 1.2789$. The corresponding cosmic times of the four benchmark epochs z_0, z_1, z_2, z_3 will be explained in Section 7.

Discussion of the Sigmoid Model Curve The model obtained in Eq. (5.20) allows us to extract several noteworthy features, summarized as follows.

- *The asymptotic future light speed.* As $t \to +35$ Gyr, the normalized light speed $C(t)$ approaches the asymptotic value $J = 1.2789$, implying that, should this sigmoid growing-$c(t)$ hold, the speed of light would continue to increase by approximately 27.89% over the next 35 Gyr to $1.2789\, c_0$.

- *A substantially extended cosmic timeline.* The CMB decoupling epoch z_0 corresponds to $t = -59.1$ Gyr, far exceeding the standard ΛCDM value of -13.8 Gyr.

- *The symmetry center of the curve.* The sigmoid function is symmetric about its inflection point at $\boldsymbol{t = -\beta} = -8.854$ Gyr, where $C(t) = 0.5J = 0.63945$.

- *A suggestive symmetry.* Remarkably, Cosmic Noon ($t_w = -17.98$ Gyr) is located near the left maximum curvature point, while the present day ($t = 0$) lies close to the right maximum curvature point, with the relation $2 \times (-8.854) = -17.708 \approx -17.98$ Gyr.

- *A philosophical reflection.* This near-symmetry, emerging from a model anchored solely by Cosmic Noon, invites reflection. The left maximum marks the epoch when star formation and black hole activity peaked, corresponding to the universe's most rapid renewal phase. The right maximum, in contrast, corresponds to the era when the universe has matured enough for intelligent life to emerge. That both extrema align so naturally with these two defining moments of cosmic history does not seem purely accidental; it hints at a resonance that might be called an aesthetics of existence.

Note: *One may construct a variant of the growing-$c(t)$ model by defining the function via the improved composite model in Eq. (5.6); however, this formulation features an overly intricate mathematical structure and thus lies beyond the scope of the present work.*

6. Ultra-fine Growing-$c(t)$ Model

In the standard ΛCDM cosmology, only cosmic redshifts can be directly observed. Precise cosmic expansion history and corresponding cosmic time are unmeasurable in practice.

All cosmic time entries stored in the `Planck18` dataset are model-derived quantities. Such calculations rely on three fundamental premises: *homogeneous and isotropic universe*, *invariant total matter and energy*, and *constant speed of light*. Cosmic ages are further solved via the Friedmann equations under these restrictions.

Under our $G(t)\, c(t)^2 = P$ postulate, $G(t)$ is time-varying, which implies that the stellar mass estimation methods used in ΛCDM contain inherent biases. A detailed discussion of this issue is deferred to Section 12.

Despite being constrained by preset theoretical assumptions and the inevitable uncertainties in model derivation, the `Planck18` data still stands as the best available reference. We therefore construct our ultra-fine growing-$c(t)$ model grounded on this dataset.

Data Extraction The `Planck18` standard redshift–cosmic time dataset (z, $t^{\triangle}$) is extracted from the Python `astropy.cosmology` library at float64 numerical precision.

The total cosmic age is approximately 13.8 Gyr. We partition the full cosmic timeline with a fixed timestep of 10^4 years (10 kyr) for uniform sampling. This yields roughly 13.8×10^5 sampled data points.

Sampling begins at the present cosmic time $t^{\triangle} = 13.79069...$Gyr, with each subsequent sample reduced by 10^4 years: the second and third sampled times equal $t^{\triangle} = 13.79068...$ and $t^{\triangle} = 13.79067...$, respectively.

The redshift of CMB epoch is $z \simeq 1090$, which defines the uppermost observational boundary. For this reason, sampling is truncated at $t^{\triangle} = 0.00037...$ and $t^{\triangle} = 0.00036...$, corresponding to $z = 1083.07$ and $z = 1100.61$, namely 37 kyr and 36 kyr after the Big Bang, in line with this physical limit.

In total, $N = 1\,379\,024$ equal-time-interval raw time-redshift pairs are extracted.

Definition of Interval Distance yr_i Following the piecewise averaging rule, the segment time distance for each time interval is defined as

$$yr_i = 1 + \frac{1}{2}\left(z_i + z_{i+1}\right) \ , \tag{6.1}$$

where $i \in (0,\ 1379022)$, and z_i denotes the redshift at the i-th sampling point. All yr_i values are computed sequentially over the full valid intervals.

Each piecewise yr_i stands for the time scaling factor derived under the growing-$c(t)$ model. For example, $yr_i = 1.5$ corresponds to an averaged redshift $\bar{z} = 0.5$ from z_i and z_{i+1}; the original 10^4 years of sampling interval is scaled to 1.5×10^4 years within the growing-$c(t)$ framework.

Cumulative Summation $\mathbf{YR_i}$ Starting from $i = 0$ and moving forward to $i = 1379022$, we accumulate the sequence yr_i following Eq. (6.2) to compute the total cosmic timescale under the growing-$c(t)$ model. To suppress floating-point round-off error arising from millions of successive additions, the Kahan summation algorithm is adopted for residual error compensation.

$$\mathrm{YR}_0 = yr_0 \ , \quad \mathrm{YR}_i = \mathrm{YR}_{i-1} + yr_i \ . \tag{6.2}$$

The resulting YR_i array represents the growing-$c(t)$ timescale quantified in units of 10^4 years. Each YR_i is multiplied by -1 to denote negative cosmic time. The formation point of the CMB epoch corresponds to -45.2 Gyr.

***C*(*t*) Normalizing and $\mathbf{YR_i}$ Downsampling** The normalized $C(t)$ follows $C(t) = 1/(1+z_t)$ from sampled redshift sequences, which produces full-resolution $C(t)$ data with $1\,379\,023$ entries.

For convenient presentation within this work, we perform downsampling by retaining one datum per 1000 raw samples. This procedure yields 1381 discrete sets of growing-$c(t)$ and corresponding time YR_i, with an effective spacing of $t^{\Delta} = 10^7$ years between neighboring data.

This dataset serves as the core database for the proposed ultra-fine growing-$c(t)$ model. Its corresponding curve is presented in following figure.

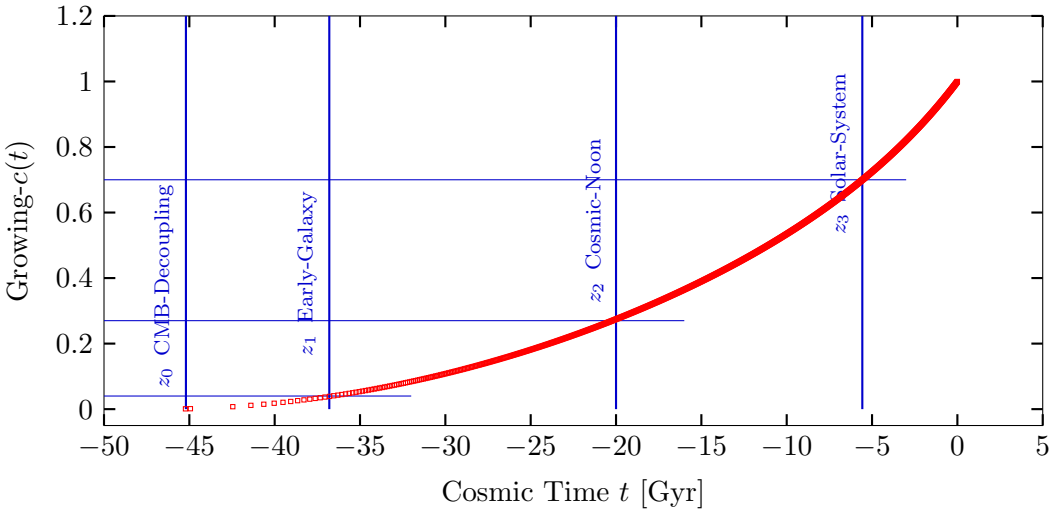


Figure 3: The C(t) curve of the ultra-fine growing-$c(t)$ model, derived from `Planck18` data. The corresponding cosmic times of the four benchmark epochs z_0, z_1, z_2, z_3 will be explained in the following section.

7. Different Physical Pictures of Growing-$c(t)$ Universe

This section depicts an extensive set of dramatically different physical pictures of cosmic evolution across various cosmic epochs.

The cosmic origin timescale is extended to $[-59.1,\ -45.2]$ Gyr, while at the CMB decoupling epoch, the light speed $c(t)$ equals 1/1100 of the present speed c_0, the Bohr radius a_{B} is only 1/1100 of its present value, and the ionization temperature of hydrogen atoms $T_{\mathrm{ion}}(t)$ amounts to merely 82.64 K.

Many physical quantities from the classical basic equations evolve alongside $c(t)$, and separate subsections will elaborate each topic sequentially.

All present-day values of fundamental physical constants used in this work (e.g., the Bohr radius $a_{\mathrm{B},0}$, the reduced Planck constant $\hbar_0$, and the proton diameter d_p) are taken from standard reference sources [4].

Cosmic Origin Timescale The CMB decoupling epoch marks the origin point of the observable universe, corresponding to a redshift satisfying $1 + z = 1100$.

Under the compact growing-$c(t)$ model defined in Eq. (5.10), $C(t) = 1/1100$ yields $t = -59.1\,\text{Gyr}$; for the ultra-fine growing-$c(t)$ model, the cosmic time reads $t = -45.2\,\text{Gyr}$ at the early CMB stage with redshift $z = 1083.07$. These results imply a gently evolving cosmos rather than a violent Big Bang scenario [5].

A preliminary assessment shows that the time span between the QGP threshold epoch ($\boldsymbol{z_\star}$) and the CMB decoupling epoch ($\boldsymbol{z_1}$) surpasses the 36-37 kyr given by the standard ΛCDM model.

The ultimate cosmic origin was postulated as the *Zero-Light-Speed Zero-Temperature* (*ZLZT*) primordial state, where both the speed of light and temperature approach zero. We infer that if the cosmic speed of light falls below 42 km/s (21 km/s or 14 km/s) and the ambient temperature stays near 2.75 K at the QGP threshold epoch ($\boldsymbol{z_\star}$), all hadronic structures fully dissociate into free quarks and gluons, forming a pure quark–gluon universe without hadrons. Detailed discussions are given in Section 9.

Cosmic Noon and $\boldsymbol{C(t)}$ Based on the compact growing-$c(t)$ model from Eq. (5.20), Cosmic Noon occurs at $t_w = -17.98\,\text{Gyr}$. For the ultra-fine growing-$c(t)$ model described in Eq. (6.2), this epoch falls at $t_w = -20.21\,\text{Gyr}$. Both timestamps differ substantially from $t^\star = -11.4\,\text{Gyr}$ predicted by the standard ΛCDM model.

At Cosmic Noon, the cosmic light speed reaches 27.03% of the present value c_0, equivalent to $1/3.7\,c_0$. Per the scaling relation $r_s \propto c^{-4}$ derived in Eq. (1.11), the Schwarzschild radius at this epoch was $3.7^4 = 187.4$ times larger than its present-day counterpart, which explains the efficient formation of black holes in this cosmic era.

Solar System Formation For the epoch of Solar System formation ($\boldsymbol{z_3}$), the `Planck18` dataset gives $t^\star = -4.67\,\text{Gyr}$ at redshift $z = 0.43$.

The compact and ultra-fine growing-$c(t)$ model yield $-7.55\,\text{Gyr}$ and $-5.57\,\text{Gyr}$, respectively. We consider the latter result more reliable, since the ultra-fine model is calibrated using the full `Planck18` dataset, whereas the compact model relies on only a single data point. The relative light speed at this epoch is 69.93%.

Data for the early galaxy formation epoch ($\boldsymbol{z_1}$) are summarized in the table below and will not be discussed further.

Table 2: $\boldsymbol{C(t)}$ and $\boldsymbol{t}$ of benchmark epochs $\boldsymbol{z_i}$

Parameter	$\boldsymbol{z_0}$	$\boldsymbol{z_1}$	$\boldsymbol{z_2}$	$\boldsymbol{z_3}$
$\boldsymbol{C(t)}$	1/1100	1/25	1/3.7	1/1.43
$\boldsymbol{C(t)}$ %	0.09%	4.00%	27.03%	69.93%
$\boldsymbol{t}_{\text{compact}}$	−59.1	−32.7	−17.83	−7.55
$\boldsymbol{t}_{\text{fine}}$	−45.2	−36.6	−20.21	−5.57
$\boldsymbol{t^\star}$	−13.8	−13.6	−11.40	−4.67

Dimensional Analysis Physics defines seven fundamental dimensional quantities, abbreviated as LMTIΘNJ. They stand for length (L), mass (M), time (T), electric current (I), thermodynamic temperature (Θ), amount of substance (N) and luminous intensity (J), respectively.

This study is built upon two core postulates listed in Section 2. The macroscopic postulate is given by $G(t)\,c(t)^2 = \boldsymbol{P}$, which indicates $G(t) \propto c(t)^{-2}$. The microscopic postulate states that the fine-structure constant $\boldsymbol{\alpha}$ is universally invariant throughout cosmic evolution.

Based on these postulates, a subset of the physical quantities from Section 1 evolve with $c(t)$. We derive their dimensions and power-law relations with $c(t)$, as summarized in the table below.

Table 3: Dimensional analysis and power-law scaling with $\boldsymbol{c(t)}$

Evolving Qty.	**Dimensional Form**	**Power-law Scaling with $c(t)$**
$G(t)$	$[G] : \mathrm{L}^3\,\mathrm{M}^{-1}\,\mathrm{T}^{-2}$	$\propto c^{-2}$
$\mu_0(t)$	$[\mu_0] : \mathrm{L}^1\,\mathrm{M}\,\mathrm{T}^{-2}\,\mathrm{I}^{-2}$	$\propto c^{+1}$
$\varepsilon_0(t)$	$[\varepsilon_0] : \mathrm{L}^{-3}\,\mathrm{M}^{-1}\,\mathrm{T}^{4}\,\mathrm{I}^{2}$	$\propto c^{-3}$
$\hbar(t)$	$[\hbar] : \mathrm{L}^2\,\mathrm{M}\,\mathrm{T}^{-1}$	$\propto c^{+2}$
$a_{\mathrm{B}}(t)$	$[a_{\mathrm{B}}] : \mathrm{L}^1$	$\propto c^{+1}$
$k_e(t)$	$[k_e] : \mathrm{L}^3\mathrm{M}\,\mathrm{T}^{-4}\,\mathrm{I}^{-2}$	$\propto c^{+3}$

The remaining seven quantities are derived from the foregoing results. Their dimensional forms and power-law scaling with $c(t)$ are presented in the following table.

Table 4: Power-law scaling of the remaining four quantities with $\boldsymbol{c(t)}$

Evolving Qty.	**Ref. Eq.**	**Power-law Scaling with $c(t)$**
$m_{\mathrm{P}}(t)$	Eq.(1.3)	$\hbar \propto c^{+2},\ c^{+1} \propto c^{+1},\ G^{-1} \propto c^{+2} \quad \Rightarrow \quad \propto \boldsymbol{c^{\frac{5}{2}}}$
$\ell_{\mathrm{P}}(t)$	Eq.(1.4)	$\hbar \propto c^{+2},\ c^{-3} \propto c^{-3},\ G^{+1} \propto c^{-2} \quad \Rightarrow \quad \propto \boldsymbol{c^{-\frac{3}{2}}}$
$\rho_{\mathrm{P}}(t)$	Eq.(1.5)	$\hbar^{-1} \propto c^{-2},\ c^{+5} \propto c^{+5},\ G^{-2} \propto c^{+4} \quad \Rightarrow \quad \propto \boldsymbol{c^{7}}$
$F_{\mathrm{P}}(t)$	Eq.(1.5)	$c^{+4} \propto c^{+4},\ G^{-1} \propto c^{+2} \quad \Rightarrow \quad \propto \boldsymbol{c^{6}}$
$T_{\mathrm{ion}}(t)$	Eq.(1.8)	$\Rightarrow \quad \propto \boldsymbol{c^{2}}$
$r_{\mathrm{s}}(t)$	Eq.(1.11)	$\Rightarrow \quad \propto \boldsymbol{c^{-4}}$
$N_{\mathrm{Dirac},1}(t)$	Eq.(1.16)	$G \propto c^{-2},\ \varepsilon_0 \propto c^{-3} \quad \Rightarrow \quad \propto \boldsymbol{c^{5}}$

Evolution of $\boldsymbol{\varepsilon_0(t)}$, $\boldsymbol{\mu_0(t)}$ $\varepsilon_0(t)$ and $\mu_0(t)$ are the core quantities governing the evolution of $c(t)$. Following $\varepsilon_0(t) \propto c(t)^{-3}$ and $\mu_0(t) \propto c(t)$, the two parameters decrease and increase respectively as $c(t)$ rises and mutually constrain one another, in accordance with the core form of Maxwell's equations, as shown in Eq. (7.1).

$$\mu_0(t)\,\varepsilon_0(t)\,c(t)^2 = 1\ . \tag{7.1}$$

The three quantities $\mu_0(t)$, $\varepsilon_0(t)$ and $c(t)$ are governed by the *Light-Electricity-Magnetism Synchronized Cosmic Dark Field* (*LEM-SCDF*), which will be fully elaborated in subsequent sections.

Evolution of the Bohr Radius $a_{\mathrm{B}}(t)$ In our framework, the Bohr radius $a_{\mathrm{B}}(t)$ of hydrogen scales linearly with $c(t)$, in contrast to the conventional assumption of a constant. At the epoch $1+z=25$, its value equals 1/25 of the present-day value. At the CMB epoch with $1+z=1100$, the Bohr radius is reduced to 1/1100 of its current magnitude.

The present-day Bohr radius is $a_{\mathrm{B},0}=5.292\times10^{-11}\,\mathrm{m}$, and the proton diameter is $d_p=1.6815\times10^{-15}\,\mathrm{m}$. Their present-day ratio is approximately 3.15×10^4. At $1+z=1100$, the ratio drops to roughly 28.6, indicating an extremely narrow electron orbit around the proton in the early universe. This ratio represents the bare minimum scale to sustain neutral hydrogen. Before the cosmic evolution reached this stage, the orbital radius was too small to form stable hydrogen atoms.

This scenario has rarely been investigated in existing studies. When the Bohr radius was merely dozens of times the proton diameter, multi-electron atoms and complex molecular bonds could hardly form, as electron shells were compressed into overlapping configurations. Additionally, the transition from plasma to neutral hydrogen during recombination was fragile and reversible; newly formed atoms were loosely bound and likely to be reionized.

The orbital scale of atoms further governs the abundance of chemical elements. In the era near CMB, the severely compacted atomic structure would have permitted only hydrogen-dominated objects to condense, with almost no complex heavy elements. Simple chemical compositions remained prevalent for a long time in the early cosmic environment.

Evolution of Coulomb parameter $k_e(t)$ As analyzed in Table 3, the Coulomb parameter $k_e(t)$ scales with the cube of cosmic light speed $c(t)$. At redshift $1+z=25$, $k_e(t)$ is $1/25^3$ of its present value, while at the CMB epoch ($1+z=1100$), it reduces to $1/1100^3$ of the current magnitude. The drastically weakened electrostatic force in the early universe led to fragile atomic bonding at the microscopic scale, consistent with the evolutionary characteristics of hydrogen atoms discussed above.

Evolution of Planck Quantities $\hbar(t)$, $m_{\mathrm{P}}(t)$, $\ell_{\mathrm{P}}(t)$, $F_{\mathrm{P}}(t)$ and $\rho_{\mathrm{P}}(t)$ As analyzed in Table 3 and Table 4, the reduced Planck constant $\hbar(t)\propto c(t)^2$, the Planck mass $m_{\mathrm{P}}(t)\propto c(t)^{5/2}$, the Planck length $\ell_{\mathrm{P}}(t)\propto c(t)^{-3/2}$, the Planck force $F_{\mathrm{P}}(t)\propto c(t)^6$, and the conventional Planck density $\rho_{\mathrm{P}}(t)$ evolves proportionally to $c(t)^7$.

In the present-day universe, the Planck mass reaches $m_{\mathrm{P},0}\approx2.2\times10^{-8}\,\mathrm{kg}\approx22\,\mu\mathrm{g}$, which is extremely massive for a microscopic scale, while the Planck length $\ell_{\mathrm{P},0}\approx1.6\times10^{-35}\,\mathrm{m}$ remains extraordinarily tiny, and the Planck density $\rho_{\mathrm{P},0}\approx5.2\times10^{96}\,\mathrm{kg/m^3}$.

Since $\rho_{\mathrm{P}}(t)\propto c(t)^7$, at the CMB epoch ($z_0$) with $1+z=1100$, we have $1100^7\approx1.95\times10^{21}$. This reduces the Planck density to $\rho_{\mathrm{P}}(t)\approx2.65\times10^{75}\,\mathrm{kg/m^3}$.

Furthermore, at the post-hadronization plasma epoch ($z_\star$), the speed of light drops to 1/7137.66 of its present-day value c_0, i.e., approximately 42 km/s. We get $7137.66^7\approx9.44\times10^{26}$. This reduces the Planck density to $\rho_{\mathrm{P}}(t)\approx5.51\times10^{69}\,\mathrm{kg/m^3}$.

Both Planck densities above are far greater than the Friedmann cosmic density $\rho_{\rm F}(t)$ defined in Eq. (1.14). We will revisit these quantities when we discuss $v_{\rm LEM}(t)$ in Eq. (8.6) later, where both $\rho_{\rm P}(t)$ and $\rho_{\rm F}(t)$ will be used.

If we eliminate G and replace it with $P = Gc^2$, the four Planck quantities in Eq. (1.3), Eq. (1.4), and Eq. (1.5) become:

$$m_P(t) = \sqrt{\frac{\hbar(t) \cdot c(t)^3}{P}}, \quad \ell_P(t) = \sqrt{\frac{\hbar(t) \cdot P}{c(t)^5}}, \quad \rho_P(t) = \frac{c(t)^9}{\hbar(t) \cdot P^2}, \quad F_{\rm P}(t) = \frac{c(t)^6}{P}\,. \tag{7.2}$$

Evolution of Boltzmann Ionization Criterion $T_{\rm ion}(t)$ As analyzed in Table 4, the ionization temperature of hydrogen atoms scales with the cosmic speed of light, following $T_{\rm ion}(t) \propto c(t)^2$. The present-day $T_{\rm ion}(t)$ is approximately $10^8\,$K, and the values of $T_{\rm ion}(t)$ at four typical cosmic epochs are given in Table 5.

Table 5: Ionization temperature $\boldsymbol{T_{\rm ion}}$ at benchmark epochs $\boldsymbol{z_i}$

Parameter	$\boldsymbol{z_0}$	$\boldsymbol{z_1}$	$\boldsymbol{z_2}$	$\boldsymbol{z_3}$
$\boldsymbol{C(t)}$	1/1100	1/25	1/3.7	1/1.43
$\boldsymbol{T_{\rm ion}}$ **(K)**	> 82.64	$> 1.60 \times 10^5$	$> 7.31 \times 10^6$	$> 4.89 \times 10^7$

According to the Boltzmann formula Eq. (1.8), if the cosmic speed of light drops to 1/7137.66 of its present value, corresponding to $c(t) = 42.03\,$km/s, the ionization temperature reduces to 1.96 K. In other words, at this light speed, any temperature exceeding 1.96 K is sufficient to dissociate neutral hydrogen atoms into plasma.

As will be further discussed in Section 9, raising the temperature from 1.96 K to 2.75 K induces quark deconfinement and hadron dissociation, transforming the plasma into a sea of free quarks and gluons. We will also conduct a detailed analysis of this phenomenon, alongside research on the initial Zero-Light-Speed and Zero-Temperature (ZLZT) cosmic state, in the same section.

Evolution of Schwarzschild Radius $r_s(t)$ As analyzed in Table 4, the radius defined in Eq. (1.11) scales as $r_s(t) \propto c(t)^{-4}$. At high redshift epochs $1 + z = 25$ and $1 + z = 7$, $r_s(t)$ is $25^4 = 390{,}625$ and $7^4 = 2{,}401$ times its present-day value, respectively. At Cosmic Noon $1 + z = 3.7$, the radius was $3.7^4 \approx 187.4$ times larger than today.

This large increase of $r_s(t)$ in the early universe had strong effects on how structures formed. With a much larger gravitational reach, early stars and compact objects could easily pull in nearby matter. As a result, they quickly cleared away the material around them, leaving little chance for planets to form. At the same time, the stronger gravity squeezed these stars more tightly, making them burn their fuel faster and evolve more quickly. This gives a new way to understand why the earliest galaxies seem to contain mostly stars, with few signs of planetary systems — a point that has not been discussed before in this context.

As cosmic $c(t)$ continues to grow toward its present value, $r_s(t)$ shrinks steadily, following $c(t)^{-4}$. This weakening of gravitational influence means that later stars are much less able to capture surrounding material. By the time the Solar System formed at $1+z \approx 1.43$, $r_s(t)$ had already dropped to about $(1.43/3.7)^4 = 22.3\%$ of its value at Cosmic Noon. The early universe was thus a place of strong gravitational activity, which gradually calmed down over time, allowing quieter planetary systems like our own to form in later epochs.

A natural extension of this picture would be to quantify the suppression of planet formation as a function of redshift. One may heuristically write

$$P_{\rm planet}(z) \;\propto\; \left(\frac{r_s(z)}{r_{\rm disk}(z)}\right)^{-\beta} \propto\; (1+z)^{4\beta}, \tag{7.3}$$

where $r_{\rm disk}(z)$ is the characteristic scale of a protoplanetary disk and $\beta > 0$ is a phenomenological index. A more rigorous derivation could be attempted from gravitational focusing or Bondi accretion physics, relating the enhanced capture cross-section to the likelihood of disk survival and subsequent planet assembly.

However, such a quantitative treatment lies beyond the scope of the present work. We include this expression solely as a heuristic indication of how the growing-$c(t)$ framework might lead to testable predictions for the redshift dependence of planetary system formation, should future observational data become available.

Evolution of the Dirac Large Number $N_{\rm Dirac,1}(t)$ As analyzed in Table 4, the Dirac larger ratio [6] in Eq. (1.16) scales as $N_{\rm Dirac,1}(t) \propto c(t)^5$. The present-day ratio is $\approx 2.27 \times 10^{39}$. At the CMB epoch with $1+z = 1100$, the large number is reduced to $(1/1100)^5$ of its current value, yielding $N_{\rm Dirac,1,CMB} \approx 1.40 \times 10^{24}$. The numerical proximity between the electric-to-gravitational force ratio and the cosmic-age-to-proton-light-crossing ratio in Eq. (1.17) is merely an accidental coincidence of the present epoch; in the early universe these numbers were not close.

Notice that such a dramatic reduction of the electromagnetic-to-gravitational force ratio at the CMB epoch is a distinctive prediction of our growing-$c(t)$ framework. A smaller $N_{\rm Dirac,1}$ implies that gravitational effects become comparatively more important in the early universe dynamics.

Re-examining the `Second` and the `Metre` Definition in SI One `Second` is defined as the duration of exactly `9,192,631,770` periods of radiation emitted from a specific energy transition of stationary cesium-133 atoms [7]. The cesium atomic frequency is a pure microscopic quantum invariant governed by the fine-structure constant and the intrinsic energy-level structure of the atom; this frequency does not rely on the speed of light. One `Metre` is defined as the path length such light travels in vacuum in one `Second`, divided by `299,792,458`.

The above definition of the `Metre` implicitly assumes that the cosmic light speed in vacuum is permanently constant. Under the present-day speed of light c_0, the wavelength corresponding to the cesium-133 transition is given by:

$$\lambda_{\rm cs} = \frac{c_0}{\nu_{\rm cs}} = \frac{299\,792\,458 \text{ m/s}}{9\,192\,631\,770 \text{ Hz}} \approx 0.032612 \text{ m} = \texttt{32.612} \text{ mm} \tag{7.4}$$

According to the above definition of the `Metre`, if the cosmic speed of light in the early universe was reduced to 75 000 km/s, that is one quarter of the present-day value of 300 000 km/s, the intrinsic radiation frequency of cesium-133 atoms remains invariant, and the corresponding wavelength $\lambda_{\rm cs}$ shrinks to one quarter of its modern magnitude, namely `8.153` mm.

Under the postulate of growing-$c(t)$, it is logically necessary to distinguish two distinct concepts: the `modern Metre` and the `atomic Metre`, denoted by $\boldsymbol{L_{\rm m}}$ and $\boldsymbol{L_{\rm a}}$ respectively. Here $L_{\rm m}$ stands for the standard metre used at present, while $L_{\rm a}$ refers to the distance travelled during $9\,192\,631\,770/299\,792\,458 \simeq 30.6633$ wave cycles of radiation from cesium-133 atoms.

In other words, $L_{\rm a}$ is defined based on fixed atomic oscillation cycles, so the total number of cycles is permanently constant. However, the actual spatial extent represented by $L_{\rm a}$ changes as the cosmic speed of light evolves following growing-$c(t)$. This conception differs markedly from conventional textbook definitions, so readers are advised to pay close attention and fully comprehend the underlying logic.

Re-examining Photon Frequency, Wavelength and Speed of Light We start with basic physical fundamentals. Constrained by the fine-structure constant and the intrinsic quantized energy gaps of atomic systems, the intrinsic frequency ν and angular frequency ω of a photon satisfy:

$$\Delta E = h\,\nu = 2\pi\hbar\,\nu = \hbar\,\omega \tag{7.5}$$

Both ΔE and $\hbar$ evolve in tandem and scale as $c(t)^2$. Consequently, the emission frequency ν of a photon is inherently constant and independent of the speed of light.

In contrast, wavelength is not an intrinsic property of photons. It varies with light speed and the refractive index of the propagation medium, following the fundamental relation for electromagnetic waves:

$$c = \lambda\,\nu \tag{7.6}$$

We adopt standard optical glass to demonstrate wavelength variation visually. For a vacuum light speed of 300 000 km/s, consider a photon with $\nu_{n0} = 1.0 \times 10^{15}$ Hz and $\lambda_{n0} = 300$ nm. Upon entering glass with a refractive index $N_{\rm glass} = 1.5$, the photon's propagation speed drops to 200 000 km/s, accompanied by a wavelength reduction to 200 nm. When exiting the glass and returning to vacuum, the wavelength returns to its original value of 300 nm.

Note: *The vacuum speed of light acts as a universal cosmic background parameter, rather than an attribute of individual photons. This common speed is governed by an as-yet-unexplored cosmic dark field, which will be identified as the LEM-SCDF field in Section 8 and Section 9.*

Within the framework of standard cosmological redshift, we examine an ancient photon with intrinsic frequency $\nu_{n1} = 1.0 \times 10^{15}$ Hz emitted at redshift $z = 3.0$, where $1 + z = 4$. Driven by cosmic expansion, its observed frequency decreases to 2.5×10^{14} Hz at the present epoch, equal to one quarter of its initial value, while its measured wavelength quadruples.

For a photon traveling across the full cosmic scale $a(t)$, standard redshift theory establishes that the total number of wave cycles along its trajectory is conserved. As the cosmic scale expands from $\frac{1}{4}a_0$ to the present scale a_0, the photon's frequency declines and its wavelength extends to adapt to the enlarged spatial scale, while the total number of wave crests and troughs remains unchanged. We term this phenomenon "redshift invariant in quantity of wave cycles".

Our growing-$c(t)$ model extends and enriches the set of physical invariants associated with cosmological redshift. At the epoch where $1+z=4.0$, the cosmic light speed is 75 000 km/s, a quarter of its present c_0 . As the cosmic spatial scale evolves from $a(t)=\frac{1}{4}\,a_0$ to a_0, light speed $c(t)$ increases synchronously. The travel time required for a photon to traverse the entire cosmic scale is conserved throughout cosmic expansion.

The dual invariant of wave cycle count and propagation time reveals profound internal harmony and the elegant aesthetics of fundamental physical laws. In essence, any photon that is emitted to travel across the whole cosmic scale $a(t)$ will maintain a fixed number of wave cycles and a constant travel time, unaffected by cosmic expansion. This intrinsic congruence is clearly manifested in our growing-$c(t)$ model.

We further analyze the ancient photon with $\nu_{n0}=1.0\times10^{15}\,\mathrm{Hz}$ emitted at $1+z=4.0$. Calibrated to the L_{m}, the cosmic light speed was 75 000 km/s at that time and the photon had an initial wavelength of 75 nm. After long-range propagation to the present, cosmological redshift lowers its frequency to 2.5×10^{14} Hz. Meanwhile, the cosmic vacuum light speed has risen to 300 000 km/s, and the observed final wavelength reaches 1200 nm, 16 times the wavelength at emission.

The total 16-fold wavelength amplification stems from two separate effects: a 4-fold contribution from cosmological redshift, and another 4-fold contribution from the monotonic increase of cosmic vacuum light speed $c(t)$.

The 16-fold wavelength shift appears counterintuitive under the assumption of a constant speed of light. Nevertheless, when interpreted using the L_{a}, we see that the photon's frequency is simply reduced to one quarter of its value at emission.

Illustrating the Materiality Behind Photon Frequency via a Metaphor Just as an individual's innate traits are fixed by parental genes, a photon's frequency at emission is solely determined by the fine-structure constant and the structure of its radiation source, independent of external cosmic environments.

After decoupling from its source, the photon's inherent frequency becomes modulated by the cosmic background. Cosmic expansion and gravitational fields of massive objects continuously alter this intrinsic frequency, predominantly producing redshift effects, much like how societal material conditions shape an individual after they enter society.

To this point, we have illustrated a series of distinctly different physical pictures describing cosmic evolution. The discussion on the Einstein $G_{\mu\nu}$ field equation, together with the Light-Electricity-Magnetism Synchronized Cosmic Dark Field (*LEM-SCDF*) proposed in this work, will be carried out in the subsequent Section 8.

8. The Revised $G_{\mu\nu}$, the Constant P, and the LEM-SCDF Field

We revisit the macroscopic postulate from Section 2: a universal time-invariant cosmic coupling constant $\boldsymbol{P}$ exists, and the time-dependent speed of light $c(t)$ satisfies the conservation law

$$G(t)\,c(t)^2 = \boldsymbol{P}\ . \tag{8.1}$$

Using this postulate, we revise Einstein's curvature field equation, and examine the physical plausibility of this $G(t)c(t)^2$ power-law relation. Furthermore, we introduce the LEM-SCDF, defined as the globally synchronized cosmic dark field of light, electricity and magnetism, as a theoretically motivated construct for this framework.

Note: *Dirac, as early as 1937 and 1938, already conceived the idea that fundamental constants might also evolve with cosmic time. He further argued that the striking coincidences among the large dimensionless numbers in physics and cosmology could be resolved if the gravitational constant G varies with the cosmic epoch $G(t^{\Delta}) \propto 1/t^{\Delta}$; and such a variation would necessarily involve a change in the speed of light $c(t)$ as well [6, 8].*

The Revised $G_{\mu\nu}$ Equation Substituting the above postulate into the Einstein curvature field equation [9] defined in Eq. (1.9), we obtain:

$$G_{\mu\nu} = \frac{8\pi G(t)}{c(t)^4} T_{\mu\nu} = \frac{8\pi P}{c(t)^6} T_{\mu\nu}\ . \tag{8.2}$$

In Eq. (8.2), $T_{\mu\nu}$ denotes the energy-momentum tensor describing mass-energy distribution. Although the full form of $T_{\mu\nu}$ for the early universe cannot be precisely determined, note that if the speed of light in the early galaxy-formation era ($1{+}z = 25$) and the CMB epoch ($1{+}z = 1100$) were respectively $1/25$ and $1/1100$ of the present-day speed of light c_0, the spacetime curvature would be extremely large, scaling as $25^6 \approx 2.4 \times 10^8$ or $1100^6 \approx 1.8 \times 10^{18}$.

Such extreme spacetime curvature in primordial epochs induced strong coupling between matter and celestial objects, which in turn demanded an exceedingly long timescale for these objects to reach relaxation equilibrium. Accordingly, the spacetime curvature (reflected by the Einstein tensor $G_{\mu\nu}$) was enormous in early epochs and exhibits a qualitative decreasing trend across successive cosmic eras: CMB epoch z_0, early galaxy-formation epoch z_1, cosmic noon epoch z_2, and the solar system formation epoch z_3.

The tensor $T_{\mu\nu}$ contains energy density, momentum flux, pressure, stress, and electromagnetic fields. On macroscopic scales, under the weak-field approximation and slow-motion approximation together with the relation $E = mc^2$, Eq. (8.2) reduces to

$$G_{\mu\nu} \approx \frac{8\pi P}{c(t)^6} \cdot M_{\mu\nu} \cdot c(t)^2 = \frac{8\pi P}{c(t)^4} M_{\mu\nu} \tag{8.3}$$

Here, $M_{\mu\nu}$ represents the density distribution of matter.

Under the postulate $G(t)\,c(t)^2 = P$, a comparison between Eq. (8.2) and Eq. (8.3) shows complete physical consistency on the right-hand sides of both equations. This indicates that the relation $\boldsymbol{Gc^2 = P}$ is ***somehow*** inherently embedded within the Einstein field equation.

Otherwise, if alternative power-law forms such as $G(t)\,c(t) = P$, $G(t)\,c(t)^3 = P$ or $G(t)\,c(t)^4 = P$ are adopted, this consistency vanishes. Although quantities associated with $G(t)$ are mathematically permissible under pure computation, they lack physically reasonable interpretation. Accordingly, such alternative power-law forms are not further discussed in this work.

Using Maxwell's equations defined in Eq. (7.1) to substitute for $c(t)^6$ in Eq. (8.2), we obtain a correlated form of the Revised $G_{\mu\nu}$ Equation involving $\mu_0(t)$ and $\varepsilon_0(t)$, as shown below:

$$G_{\mu\nu} = 8\pi\, P \cdot \mu_0(t)^3 \cdot \varepsilon_0(t)^3 \cdot T_{\mu\nu} \tag{8.4}$$

Theoretical Insight of the Revised $G_{\mu\nu}$ Equation The major part of cosmic mass and energy originates from the strong interaction. The binding energy between quarks and gluons creates most mass of visible matter, so the energy-momentum tensor $T_{\mu\nu}$ essentially reflects the macroscopic distribution of strong-interaction effects in the universe.

Einstein's original field equation forms a purely one-way geometric response model. All variations in the mass-energy distribution described by $T_{\mu\nu}$ are balanced solely via spacetime curvature $G_{\mu\nu}$, with no extra physical degree of freedom available to share the dynamical burden.

In contrast, our revised framework establishes a closed-loop regulation system. The spacetime curvature $G_{\mu\nu}$ cooperates with the time-varying speed of light $c(t)$ to jointly respond and balance the dynamic variation of $T_{\mu\nu}$.

Since $c(t)$ is determined by vacuum electromagnetic parameters $\mu_0(t)$ and $\varepsilon_0(t)$, the entire cosmic system effectively connects microscopic strong-interaction mass sources, large-scale spacetime geometric effects, and underlying vacuum electromagnetic rules. All physical variables in this dynamic process are strictly constrained by the invariant cosmic constant P.

Accordingly, the revised $G_{\mu\nu}$ equation no longer belongs to a pure geometric theory of gravity. It forms a unified cosmic dynamic framework grounded on vacuum electromagnetic fundamentals, adopts spacetime curvature and time-varying light speed as dual regulatory channels, and takes the strong-interaction-dominated mass-energy distribution as the core source term.

The Intrinsic Meaning of Constant P P is termed the cosmic coupling constant, as it characterizes the fundamental coupling among cosmic mass-energy distribution, spacetime curvature, the time-varying speed of light $c(t)$, and the time-varying vacuum permeability $\mu_0(t)$ and vacuum permittivity $\varepsilon_0(t)$.

By manipulating Eq. (8.4), Eq. (8.2) and Eq. (8.3), we obtain:

$$\boldsymbol{P} = \frac{G_{\mu\nu}}{8\pi\,\mu_0(t)^3 \cdot \varepsilon_0(t)^3 \cdot T_{\mu\nu}} = \frac{c(t)^6 \cdot G_{\mu\nu}}{8\pi \cdot T_{\mu\nu}} \approx \frac{c(t)^4 \cdot G_{\mu\nu}}{8\pi \cdot M_{\mu\nu}} \tag{8.5}$$

▷▷▷ Historically, Einstein held an ambivalent attitude toward gravitational constant. Although general relativity attributes gravitational phenomena entirely to spacetime curvature, rendering G no longer a fundamental independent quantity, Einstein nevertheless retained G in his field equations. This choice was primarily intended to ensure the equation reduces to Newtonian gravitation under weak-field and slow-motion limits, rather than by rigorous theoretical necessity.

For roughly a decade after 1915, Einstein and most contemporary physicists adhered to the static cosmological paradigm — a view that remained unchallenged until Hubble's 1929 discovery of the expanding universe. This intellectual framework made it virtually inconceivable to entertain time-varying gravity or light speed, yet the invariant relation $Gc^2 = P$ is somehow inherently embedded within this equation. It was not until 1937 that Dirac began to question whether G might evolve with cosmic time $G(t^{\Delta}) \propto 1/t^{\Delta}$. $\triangleleft\triangleleft\triangleleft$

The LEM-SCDF Field Within the growing-$c(t)$ framework, the vacuum electromagnetic parameters c, ε_0, and μ_0 are no longer universal constants. Following the standard postulation paradigm in theoretical physics, this work introduces the Light-Electricity-Magnetism Synchronized Cosmic Dark Field (LEM-SCDF) as a fundamental cosmic background field.

This study focuses on the LEM–SCDF dark field throughout cosmic evolution from the post-hadronization plasma epoch, across the CMB era, and up to the present universe. All QCD-governed quark-gluon dynamics and hadronization processes are excluded from the current discussion.

The core function of the LEM–SCDF dark field is to achieve global synchronization of the time-varying fundamental parameters $c(t)$, $\varepsilon_0(t)$, and $\mu_0(t)$ across the entire cosmic space, while maintaining their strict spatial uniformity.

We denote by $\boldsymbol{v_{\mathrm{LEM}}(t)}$ the propagation speed of the LEM-SCDF dark field, which satisfies $v_{\mathrm{LEM}}(t) \gg c(t)$. We define the invariant relation as

$$v_{\mathrm{LEM}}(t) \cdot \rho_{\mathrm{F}}(t) = c(t) \cdot \rho_{\mathrm{P}}(t) \ , \tag{8.6}$$

or equivalently,

$$v_{\mathrm{LEM}}(t) = c(t) \cdot \frac{\rho_{\mathrm{P}}(t)}{\rho_{\mathrm{F}}(t)} \ . \tag{8.7}$$

Here, $\rho_{\mathrm{F}}(t)$ denotes the mean cosmic density that appears in the first and second Friedmann equations, Eq. (1.14) and Eq. (1.15), and serves as the core density term governing cosmic expansion. The quantity $\rho_{\mathrm{P}}(t)$ is the Planck density discussed in Section 7, it is the upper limiting density imposed by quantum gravity, an inherent property of spacetime itself.

On the right-hand side of Eq. (8.6), $c(t) \cdot \rho_{\mathrm{P}}(t)$, the term $c(t)$ is the baseline propagation speed for electromagnetic and gravitational interactions in local spacetime, while $\rho_{\mathrm{P}}(t)$ is the maximum mass density that spacetime can sustain under quantum-gravitational constraints. Their product thus gives the intrinsic upper limit of interaction transmission per unit volume.

On the left-hand side, $v_{\mathrm{LEM}}(t) \cdot \rho_{\mathrm{F}}(t)$, the Friedmann mean density $\rho_{\mathrm{F}}(t)$ represents the diluted actual distribution of matter in the expanding universe, and $v_{\mathrm{LEM}}(t)$ is the characteristic propagation speed of the LEM–SCDF dark field.

This symmetric product structure establishes a dual correspondence between the microscopic quantum spacetime limit and the macroscopic large-scale cosmic evolution, and it holds for arbitrary cosmic time t. The equality of these two products maintains a permanent balance between the quantum-scale bound of spacetime and the coordination requirements of the entire macroscopic cosmos. This definition is mathematically well-behaved, with no singularities or non-differentiable points. It also has formal elegance and a clear physical picture.

Since $\rho_{\mathrm{P}}(t)$ is the absolute upper density limit allowed by quantum gravity, and since, apart from the initial singular state of cosmic expansion (which is a logical idealization), we have $\rho_{\mathrm{P}}(t) \gg \rho_{\mathrm{F}}(t)$ at all subsequent evolutionary stages, this inequality directly implies $v_{\mathrm{LEM}}(t) \gg c(t)$ at any given time.

The $v_{\mathrm{LEM}}(t)$ exceeds the local vacuum speed of light by many orders of magnitude, so the propagation time delay of the dark field remains negligibly small across all observable cosmic scales. This feature naturally produces nearly instantaneous global synchronization, which underpins the core invariant $G(t)\,c(t)^2 = \text{constant}$ established in this work.

The global synchronization dominated by the LEM–SCDF dark field requires no material medium, involves no transmission of energy or physical signals, and implies no causal correlation between emitting and receiving particles. Its global coordination is essentially a state transition of the cosmic background itself. Its superluminal propagation speed therefore does not violate the local light-cone causality constraints of special relativity.

Note: *The speed* $\boldsymbol{v_{\mathrm{LEM}}(t)}$ *is precisely the speed that governs quantum entanglement, a fundamental nonlocal correlation rooted in the quantum vacuum. This resolves the century-old paradox that Einstein famously dismissed as "spooky action at a distance" in his debates with Bohr. What Einstein and Bohr fiercely debated as a metaphysical puzzle is now given a physical ontology: the LEM-SCDF field itself is the medium through which quantum entanglement is globally synchronized across cosmic scales.*

All mass and energy in the cosmos collectively participate in shaping the evolution of these LEM fundamental parameters. Consequently, the LEM-SCDF dark field provides a unified underlying physical framework that consistently integrates Maxwell's electromagnetism, Yang–Mills gauge field theory, and Einstein's general relativity.

Note: *Standard Maxwell's vacuum parameters* c, ε_0 *and* μ_0 *are measured constants obtained from present-day clean vacuum experiments. There is no proof that these fixed values apply to all eras of cosmic evolution. In the LEM-SCDF framework, we discard static vacuum electromagnetic constants* c, ε_0 *and* μ_0, *and instead use time-varying global average background parameters to describe the cosmic electromagnetic substrate. The clean modern-style vacuum corresponds to an ideal electromagnetic environment and cannot be treated as the general cosmic state. Such ideal vacuum conditions were absent at least in two key cosmic periods: the post-hadronization plasma phase prior to CMB decoupling, and the reionization epoch up to redshift* $1 + z = 25$, *when the whole universe was filled with ionized plasma. It is unreasonable to directly apply today's vacuum electromagnetic standards to these early cosmic stages.*

Alternative Approaches to UFT Puzzles Traditional Unified Field Theory (UFT) research has long been hindered by two fundamental obstacles: the quantization of gravity and the photon-scattering problem.

The quantization of gravity poses the first obstacle: general relativity exhibits essential structural incompatibility with standard Yang–Mills gauge field frameworks, rendering conventional quantization schemes infeasible.

Our revised $G_{\mu\nu}$ equation abandons the traditional constant gravitational parameter G and adopts a time-evolving $c(t)$ variable. By shifting to LEM field quantization, this modified cur-

vature formulation may provide a viable alternative for circumventing the gravitational quantization dilemma.

As a tentative direction for further exploration, we note that by combining Eq. (8.5) and Eq. (1.6), a formal connection can be drawn between the macroscopic gravitational sector and the microscopic quantum domain:

$$\boldsymbol{P} = \frac{G_{\mu\nu}}{8\pi\,\mu_0(t)^3 \cdot \varepsilon_0(t)^3 \cdot T_{\mu\nu}} = \frac{c(t)^6 \cdot G_{\mu\nu}}{8\pi \cdot T_{\mu\nu}} = \frac{\hbar(t)^4 \cdot \rho_{\rm P}(t)}{m_{\rm P}(t)^6} \quad . \tag{8.8}$$

The above equation brings together the languages of general relativity and quantum mechanics in one place. Its left-hand side speaks in terms of spacetime curvature and mass-energy; its right-hand side, in terms of Planck-scale quantities.

▷▷▷ In the years following the completion of general relativity (1915–1920), Einstein redefined spacetime as a physical continuum endowed with intrinsic properties. His 1920 lecture on the new ether at the University of Leiden, "Ether and the Theory of Relativity" [10] — a classic text that has recently become available in a modern translation, "Einstein: On Ether and Relativity" [11] — explicitly argues against the notion of empty space. He held that:

- spacetime is not an empty geometric background; geometry itself is the physical attribute of spacetime, with curvature being an inherent material property of spacetime itself;
- the matter-energy tensor alters the physical properties of spacetime (curvature and density), and in turn, these spacetime properties govern the trajectories of matter and light;
- there exists no "*vacuum*" devoid of material attributes — the *vacuum* is a continuous medium with a mutable physical structure;
- according to general relativity, space without ether is unthinkable: in such a space, not only would the propagation of light be impossible, but measuring-rods and clocks — the very standards of space and time — could have no physical existence, and thus no space-time intervals in the physical sense could be defined.

Reflecting on the relationship between the gravitational and electromagnetic fields, Einstein observed that while the gravitational field is inseparable from space itself, "a part of space may very well be imagined without an electromagnetic field; thus in contrast with the gravitational field, the electromagnetic field seems to be only a secondary phenomenon." He then articulated the aspiration that "*it would be a great advance if we could succeed in comprehending the gravitational field and the electromagnetic field together as one unified conformation; then the contrast between ether and matter would fade away.*"

I agree with Einstein's aspiration, but not with his characterization of the electromagnetic field as secondary. In our envisioned physical picture of LEM-SCDF, the two fields are intrinsically unified at the quantum level, and the quantized LEM-SCDF itself already accounts for spacetime curvature, which suggests that electromagnetism may be the more fundamental partner in this synthesis; and that the LEM-SCDF itself is unified with a global quantum field theory at the Planck scale. A detailed exposition of the underlying physics, however, lies beyond the scope of the present discussion, as the framework remains incomplete and requires further development. ◁◁◁

The second critical issue is the photon-scattering problem. Standard ΛCDM cosmology assumes a hot primordial universe with constant present-day c, which produces intense photon scattering and obstructs early particle assembly and hadronization. The cosmic transition from opaque plasma to transparent medium is governed by coupled scattering processes such as Thomson scattering on free electrons and Rayleigh scattering on neutral hydrogen, as documented in the existing literature [12].

To resolve the horizon and flatness puzzles, Magueijo introduced an early-time VSL (variable speed of light) model [13] in which the speed of light in the early universe is assumed to be extraordinarily larger than its present-day c. However, this ansatz amplifies photon interaction rates and mean free paths, worsening the scattering barrier to hadronization and neutral atom formation — the very problem that a successful cosmology must resolve.

Within the growing-$c(t)$ framework, this issue is addressed via the postulated *ZLZT* (Zero -Light-Speed and Zero-Temperature) primordial universe, where both light speed and temperature asymptotically approach zero. This initial condition removes the intense photon scattering inherent to standard cosmology and conventional VSL theories.

The following Section 9 further discusses the physical implications of the ZLZT primordial postulate. It investigates cosmic evolution starting from the post-hadronization plasma stage through to the CMB epoch, focusing on the physical generation of *Arche-Particle-Substance* and the gradual growth of light speed from an initial near-zero state to $c(t) = 1/1100$ at the CMB epoch.

9. The ZLZT Primordial State, Confinement Generates Substance, and $c(t)$ Evolution

Under the assumption of a constant speed of light, the SI definition of mass establishes a tight connection with the speed of light and energy. In this section, we introduce $\boldsymbol{M}_{\mathbf{aps}}$, standing for *Arche-Particle-Substance*, to describe pure substance that is decoupled from the speed of light.

Relying on known static QCD results for bound hadron states, we restructure the conventional energy–mass framework based on the M_{aps} scale, and propose a distinctive viewpoint: *Confinement Generates Substance*. This idea differs fundamentally from the conventional interpretation that confinement corresponds to energy accumulation.

This section studies how cosmic substance forms, as well as the gradual rise of the cosmic light speed $c(t)$ from the post-hadronization plasma phase through the CMB era. This work does not explore the intrinsic QCD-driven dynamical mechanism of hadronization.

Note: *In our model, substance, namely Arche-Particle-Substance (M_{aps}), is an intrinsic physical entity independent of the speed of light.*

A Critical Note on Mass Definition in the New SI System The 2018–2019 revision of the International System of Units (SI) redefined the kilogram based on the Planck constant h, aiming to build a complete unit system upon a minimal set of quantized fundamental physical constants. This revision substantially enhances the long-term stability and measurement accuracy of metrological standards.

Nevertheless, this framework blurs the conceptual boundary between intrinsic matter and energy. By anchoring the definition of mass entirely within the energy framework, the official standard treats mass merely as a derivative manifestation of energy. By contrast, the concept of rest mass in traditional physics education provides clear physical intuition.

From the author's perspective, it is improper to define mass merely as an *energy-equivalent form* at the fundamental level simply to pursue *minimalism* in dimensional foundations. Instead, a more reasonable approach is to maintain clear distinctions at the conceptual level of the SI standard, while adopting the current quantized scheme at the high-precision measurement level.

Arche-Particle-Substance The term *arche* derives from Ancient Greek, signifying origin, commencement and fundamental principle. Combined with *particle substance*, M_{aps} denotes the primordial fundamental substance of cosmic particles.

In modern physics, the electron rest property is commonly quoted in MeV, a unit for rest energy instead of intrinsic mass. The genuine rest inertial substance is derived by dividing this energy by c^2.

The proposed quantity M_{aps} takes the electron rest inertial mass M_{e} as its calibration reference, clearly distinguishing the concept of substance from energy conversion.

Hadronization Confinement Energy in QCD In quantum chromodynamics (QCD), mass is described in terms of energy defined by the constant speed of light. The proton rest energy is approximately $m_p c^2 \approx 938.3$ MeV, which consists of four components.

$$E_p = m_p c^2 = (2m_u + m_d)c^2 + T_{\text{quark}} + E_{\text{gluon}} + V_{\text{potential}} \ . \tag{9.1}$$

The first component is the contribution from two up quarks and one down quark. The combined rest energy of its three valence quarks (uud) is

$$(2m_u + m_d)c^2 \approx 8.8 \text{ MeV} \ . \tag{9.2}$$

This quark rest energy accounts for less than 1% of the proton's total energy. In QCD, the remaining nearly 99% originates from confinement, namely hadronization energy. Therefore, the proton mass is predominantly contributed by confined field energy rather than quark rest mass. The detailed energy composition is expressed as:

- T_{quark} denotes the kinetic energy of the three confined quarks. Analogous to three billiard balls moving vigorously inside the hadron at relativistic speeds, this term is approximately 600 MeV.
- E_{gluon} stands for the gluon field energy within the hadron. Gluons act like highly viscous glue, moving rapidly inside the hadron and maintaining the volume of the hadronic bag, with an energy of approximately 340 MeV.

- $V_{\text{potential}}$ refers to the perturbative potential energy generated by first-order gluon exchange, which is approximately -25 MeV.

In the above QCD model, the total energy of a proton E_p is 938.3 MeV, and the energy of a free electron E_e is 0.511 MeV. The value of the former is **1836.15** times that of the latter, which is the well-known mass ratio between a proton and an electron.

Confinement Generates Substance In QCD, confinement traps energy [14]; in this work, *Confinement Generates Substance*. This marks a fundamental distinction. Under the assumption of a constant speed of light, energy and substance can be regarded as nearly interchangeable, whereas in a variable-c framework, the two concepts must be strictly differentiated.

Confinement is not a simple process of energy storage. Instead, the confinement process itself is a *substance-forming process*. The majority of cosmic substance originates from hadronization, making confinement the dominant mechanism for the generation of cosmic substance.

High-energy experiments conducted at CERN, Fermilab, BNL and SLAC investigate protons in highly excited states. The observed behaviors of quarks and gluons are energy-driven responses under extreme excitation, and do not necessarily reflect the genuine ground-state substance structure inside hadrons.

The internal structure of a proton has never been directly observed. Accordingly, the energy signatures detected in particle collisions cannot be equated with the intrinsic substance nature residing within hadrons.

Note: *In this model, hadronization is equivalent to materialization. Hadrons function as cosmic real estate — fixed substance assets — while the speed of light acts as the cosmic energy–substance exchange rate. The bulk of cosmic substance formed during early hadronization at an extremely low $c(t)$, corresponding to this low exchange rate.*

QCD attributes cosmic mass to energy freezing at ultra-high temperatures under a constant speed of light; by contrast, this work traces cosmic substance to confinement-driven substance generation under the low early values of c. These two physical pictures differ fundamentally.

Once formed, confined substance M_{aps} permanently remains as substance rather than energy. Any conversion into energy follows the contemporary exchange rate $c(t)^2$, so the energy equivalent of a given quantity of substance rises as $c(t)$ increases. It therefore serves as a core postulate of this model that M_{aps} is conserved, while energy is not.

For the proton, QCD quantifies confinement as an energy of approximately 938.3 MeV. By comparison, this work describes the confined content as roughly **1836.15** electron-M_{aps} units of substance, which directly connects the confinement effect to substance scale.

Photon Scattering and Primordial Temperature Standard QCD cosmology assumes a hot primordial universe with two unsolved flaws: intense photon scattering blocks hadronization, and cosmic expansion cannot account for uniform early thermal distribution.

The growing-$c(t)$ ZLZT state begins with near-zero temperature and light speed, erasing strong primordial photon scattering and removing obstacles to hadronization.

Hadronization captures surrounding energy to form M_{aps} substance. This energy-consuming process cools the cosmos and yields a naturally homogeneous early thermal field.

Temperature and Light-Speed Thresholds in the ZLZT Ansatz As the speed of light c is universally treated as a constant, standard quantum chromodynamics (QCD) studies on the hadron-to-quark-gluon plasma (QGP) transition only examine the temperature dependence of the deconfinement critical point. No existing literature addresses how a variable light speed would alter this critical threshold.

Hydrogen ionization is a QED-governed two-body dissociation process that obeys the Bohr scaling law, yielding a quadratic light-speed dependence $\boldsymbol{T_{\text{ion}} \propto c^2}$ for the critical ionization temperature.

To construct a self-consistent variable-$c(t)$ cosmological model, we adopt a simple power-law relation to distinguish QED and QCD phase transitions.

Within hadrons, quarks and virtual gluons exhibit strong multi-body collective interactions. We therefore hypothesize the QCD deconfinement temperature follows a cubic light-speed scaling $\boldsymbol{T_{\text{QGP}} \propto c^3}$. This relation is just a phenomenological assumption of the ZLZT framework, not a standard result of fixed-c QCD.

Under this assumption, if the speed of light falls to 1/7137.66 of its present value, corresponding to $c(t) \approx 42\,\text{km/s}$, an extremely low value approaching zero, the hydrogen ionization temperature scales down to 1.96 K. In other words, at this reduced light speed threshold, any temperature above 1.96 K will dissociate neutral hydrogen atoms into electromagnetic plasma.

If we apply this cubic scaling rule to QCD deconfinement, the current QGP transition temperature of $10^{12}\,\text{K}$ would scale down by a factor of $1/7137.66^3$ during the early ZLZT stage. This results in a residual temperature of about 2.75 K.

From this sequential inference, if the cosmic speed of light drops below 42 km/s (or 21 km/s, 14 km/s) while the surrounding temperature remains close to 2.75 K at the QGP threshold epoch $\boldsymbol{z_\star}$, all hadronic structures dissociate completely into free quarks and gluons, yielding a pure quark–gluon universe free of bound hadrons.

This constitutes our inference on the speed-of-light and temperature thresholds of the primordial Zero-Light-Speed and Zero-Temperature (ZLZT) state. When the temperature is fixed at 2.75 K and the speed of light rises from 21 km/s to 42 km/s or 63 km/s, the hadronization process accelerates gradually, and more frequent transitions from quark-gluon plasma to hadrons take place. Correspondingly, the Confinement-Generates-Substance process becomes increasingly active.

The Inapplicability of LEM-SCDF to the QGP Epoch Maxwell's equations only describe electromagnetic (U(1)) interactions and carry a limited scope of validity. In contrast, the quark-gluon plasma (QGP) epoch is governed predominantly by the strong force (SU(3) color interactions). This draws a distinct physical boundary: classical electrodynamics fails to fully characterize the behaviors arising within QGP.

Furthermore, all existing theoretical and experimental investigations of QGP adopt the assumption of a constant vacuum light speed c_0, which stands in stark conflict with the variable-c premise laid out in LEM-SCDF.

The LEM-SCDF framework is postulated to hold from the post-hadronization plasma epoch through the CMB decoupling epoch up to the present day, yet we explicitly exclude QGP and the hadronization transition from the domain of validity of LEM-SCDF.

Substance Clustering Shapes $c(t)$ Dynamics This subsection puts forward a tentative directional conjecture: prior to the CMB epoch, the temporal growth of the cosmic speed of light $c(t)$ originates from the local clustering of cosmic substance.

After hadronization completes, the originally homogeneous and dispersed substance begins to accumulate within localized spatial regions, while bound hadronic substance becomes progressively denser. This is the substance clustering process we address here.

This difference in structure breaks the early homogeneous distribution of cosmic substance. It changes the overall electromagnetic background controlled by the LEM-SCDF field.

As substance continuously accumulates into dense localized structures, the spatial inhomogeneity of cosmic substance distribution intensifies. The background electromagnetic parameters coordinated by LEM-SCDF evolve synchronously, which further drives the gradual growth of cosmic $c(t)$.

This forms a self-strengthening feedback loop. To provide an intuitive mathematical reference for this line of reasoning, we employ statistical methods and construct a simplified two-dimensional model to quantify the degree of substance clustering.

Consider a two-dimensional square region Ω with total area A. Let $\rho(x, y)$ denote the areal density distribution over the plane, with the total substance conserved as

$$S = \iint_{\Omega} \rho(x, y)\, \mathrm{d}x\mathrm{d}y = \text{constant} \ . \tag{9.3}$$

The global mean density is given by

$$\bar{\rho} = \frac{S}{A} \ . \tag{9.4}$$

For a perfectly uniform distribution, $\rho(x, y) \equiv \bar{\rho}$ holds everywhere inside Ω, which corresponds to the early fully homogeneous post-hadronization plasma state we postulate.

When substance gradually aggregates into localized clumps while the total amount of substance remains unchanged, the density field becomes inhomogeneous, with certain regions denser and others more rarefied.

To quantify the magnitude of spatial inhomogeneity and dispersion, we introduce the variance of the substance density field:

$$\sigma_{\rho}^2 = \frac{1}{A} \iint_{\Omega} \left[\rho(x, y) - \bar{\rho}\right]^2 \mathrm{d}x\mathrm{d}y \ . \tag{9.5}$$

In the early uniform state, $\sigma_{\rho}^2 = 0$.

As substance clusters and the distribution grows more uneven, σ_ρ^2 rises monotonically. We define a dimensionless variation coefficient as

$$C_\rho = \frac{\sigma_\rho}{\bar{\rho}} = \frac{1}{\bar{\rho}} \sqrt{\frac{1}{A} \iint_\Omega \left[\rho(x,y) - \bar{\rho}\right]^2 \mathrm{d}x\mathrm{d}y} \ . \tag{9.6}$$

To enable cross-comparison across distinct cosmic evolutionary stages. Larger values of C_ρ correspond to stronger spatial differentiation of substance, consistent with the rising trend of $c(t)$ stated in our conjecture.

From the perspective of information theory and statistical entropy, we express the normalized probability density as

$$p(x,y) = \frac{\rho(x,y)}{S} \ , \tag{9.7}$$

which obeys the normalization condition $\iint_\Omega p(x,y)\,\mathrm{d}x\mathrm{d}y = 1$. The information entropy characterizing the ordering of the substance distribution reads

$$H = -\iint_\Omega p(x,y)\,\ln p(x,y)\,\mathrm{d}x\mathrm{d}y \ . \tag{9.8}$$

With fixed total substance and fixed geometric boundaries, a perfectly uniform distribution yields the maximum entropy $H_{\max} = \ln A$. As substance assembles into clustered structures, the system becomes more ordered and entropy decreases. We accordingly define the entropy reduction

$$\Delta H = H_{\max} - H \ , \tag{9.9}$$

as a direct metric for substance aggregation and spatial inhomogeneity.

Taken together, the density variance σ_ρ^2 and entropy reduction ΔH constitute a consistent dual statistical description of substance dispersion and clustering within the plasma epoch.

This model only serves as a heuristic auxiliary illustration to clarify the correlation between substance inhomogeneity and the evolutionary trend of $c(t)$. The full mathematical construction and rigorous physical proof linking substance clustering to the background electromagnetic parameters of LEM-SCDF are left for future research.

Note: *In this picture, the cosmic speed of light $c(t)$ is a statistical collective property of the universe — just as gas temperature is not a property of any single molecule, but a statistical average of their collective motion. Every star, every milligram of substance, and every joule of energy contributes, through its electromagnetic configuration, to the global response of the LEM-SCDF background. As matter condenses, the statistical distribution of these sources shifts, and $c(t)$ evolves accordingly. In short, $c(t)$ is to the cosmos what gas temperature is to a gas: a macroscopic statistical quantity, not a metaphysical constant.*

The Dawn of Neutral Atoms: Marching toward CMB Decoupling As the local condensation of cosmic substance continues, the speed of light $c(t)$ rises steadily. This increasing light speed directly strengthens the Coulomb interaction between protons and electrons. Consequently, the characteristic orbital radius determined jointly by $c(t)$ and the fine-structure

constant gradually expands from an extremely tiny initial value until it becomes large enough to accommodate electrons orbiting protons, reaching roughly 1/1100 of the present-day Bohr radius derived in Section 7.

Under this enhanced electromagnetic binding force, electrons finally gain the physical conditions required to stably orbit around protons. Although the forming atomic structures are still prone to re-ionization at this early stage, the probability of maintaining stable bound states increases continuously as $c(t)$ grows. As more electrons successfully bind to protons and primitive hydrogen atoms accumulate in large quantities, the universe begins to transition away from its plasma-dominated state.

During the cosmic evolution from the post-hadronization plasma epoch ($\boldsymbol{z_\star}$) to the CMB decoupling epoch ($\boldsymbol{z_0}$), the speed of light gradually increases from below 42 km/s to 1/1100 (272 km/s) of the present-day c_0. Under the assumption of such an extremely low light speed in the early universe, this evolutionary process may span 4–5 Gyr, which is vastly longer than the 36–37 kyr predicted by the standard ΛCDM model. This span is obtained by a simple power-law estimation from the $c(t)$ curve, covering the rise from 1/7137.66 to 1/1100 of c_0.

This continuous formation of neutral hydrogen ultimately guides cosmic evolution onward toward the epoch of CMB decoupling, opening a brand-new cosmic realm governed by Einstein's field equations.

10. Cosmological Redshift: An Unanswered Causal Gap and Its Physical Closure

For a full century, modern cosmology has left a fundamental logical gap unaddressed. In this section, we confront this gap directly.

Our Challenge We begin with a concrete result from Section 7. There, we re-examined the behavior of photon frequency, wavelength and the speed of light. Consider an ancient photon with $\nu_{n0} = 1.0 \times 10^{15}\,\mathrm{Hz}$, emitted at redshift $1 + z = 4.0$. After nearly ten billion years of propagation through intergalactic space, cosmological redshift has reduced its frequency to $2.5 \times 10^{14}\,\mathrm{Hz}$ at the present epoch.

We put a direct question to the cosmologist: *"The photon has left its host galaxy and travelled through nearly empty space for ten billion years. Why does it obediently, strictly following the rules, know that it must redshift as the universe expands? Who notified the photon? How was the photon notified?"*

The standard answers are familiar. One says: space itself expands, stretching the photon's wavelength, just as waves drawn on a balloon elongate when the balloon is inflated. Another says: in the FRW metric, the scale factor $a(t)$ enters the null geodesic equation, and the frequency naturally scales as $a(t)^{-1}$; the calculation gives the redshift, so no further mechanism is required.

Neither answer, however, directly addresses the physical mechanism behind photon redshift. Both are either geometric analogies or mathematical re-statements that effectively mask the underlying causal question.

We return to our challenge. The logical chain is remarkably clear:

- **Premise_1**: The photon has long since departed from all material sources — its emitting galaxy, massive bodies, and local gravitational fields.
- **Premise_2**: During its flight, while the cosmic scale expands, the photon's frequency redshifts strictly and proportionally.
- **Premise_3**: If the photon in flight has no remaining connection to the rest of the universe, by what means does it *know* that the universe has expanded? By what means does it *adjust* its frequency accordingly?
- **Premise_4**: If the photon remains in continuous contact with the universe throughout its flight, so as to instantly receive the expansion information and respond with redshift, then *must not* the propagation speed of this contact mechanism be vastly greater than the speed of light itself?
- **Premise_5**: Is the message carried by this contact a direct instruction about the "cosmic expansion scale"? That seems impossible. The geometric quantity of scale expansion has no inherent physical connection to the electromagnetic oscillations of a photon.
- **Premise_6**: Could the contact instead deliver individual frequency-adjustment commands to each photon separately? That is equally impossible. Photons across the cosmos possess a vast range of different frequencies; there is no universal command that could tell every photon exactly how much to shift.
- **Premise_7**: With all other possibilities excluded, *do you not agree* that only one self-consistent resolution remains? The underlying signal transmitted by a global field can only be the intrinsic property of light itself — the global synchronized variation of the vacuum light speed $c(t)$. No multiple alternative answers exist.

The existing logical gap, in our view, constitutes a genuine ***causal deficit*** in the standard redshift interpretation.

Nor are these concerns new. The physical mechanism underlying cosmological redshift has been questioned before: Feynman(1960) [15] noted that it is not a Doppler shift but a change occurring to the photons themselves during their journey; Davies(1995) [16] further identified a persistent causal deficit, arguing that attributing redshift to spatial expansion alone replaces physical mechanism with geometric description.

A Method Taught by History The history of physics teaches us: inferring a causal gap in existing theory from observational appearances, and thereby revealing an as-yet-undiscovered global physical entity, is a powerful and productive method of thought — one that has succeeded repeatedly at critical moments in the history of science.

Einstein, confronted with the striking equivalence between inertial and gravitational mass, refused to treat it as coincidence; he concluded that gravity cannot be an action-at-a-distance

force, but must be a manifestation of spacetime geometry itself — and thus discovered general relativity.

Oersted first demonstrated that electric currents generate magnetic fields. Drawing on this symmetry, Faraday conjectured the reverse effect: a time-varying magnetic field should induce an electric field, and he verified this phenomenon experimentally to establish the law of electromagnetic induction.

In both cases, the crucial step was not fitting the data, but recognizing that the existing framework lacked a physical agent — and then demanding that the agent must exist.

The Only Causally Sound Conclusion We apply the same reasoning here. The only causally sound conclusion is this: There must exist a mechanism — propagating far faster than light and covering the entire universe — that continuously notifies and modulates the photon throughout its entire journey.

The cosmos must be pervaded by a global, extremely fast synchronized field, to which every photon remains coupled along its entire path of propagation. This field responds to the global dilution of the cosmic average density and to the universal evolution of the speed of light, continuously adjusting the photon's frequency in accordance with the expanding background.

In other words, the physical interpretation of cosmological redshift requires the existence of a universal background field with propagation speed far exceeding that of light. This is precisely the core function that the LEM–SCDF dark field fulfills.

The LEM propagation speed definition in Eq. (8.6) is not unique; the Planck and Friedmann densities are adopted as heuristic scaling factors to define a sufficiently large propagation speed for the LEM-SCDF field. Other mathematically self-consistent forms could also be constructed. What is not optional, however, is the physical necessity of the LEM-SCDF field itself: its propagation speed must be vastly greater than $c(t)$.

This is a logical and physical necessity: a photon, long after leaving all local sources, continues to obey a frequency evolution that cannot be explained by local interactions alone. Behind this observed behavior, there must exist a physical mechanism that propagates far faster than light and maintains continuous contact with the universe as a whole.

Cosmic Redshift is Just the Cosmic Light-Speed Expansion From the above challenges we have posed, to the conclusion we have reached, one *unavoidable* truth emerges: cosmic redshift is just the expansion of the global speed of light $c(t)$!

For an ancient photon emitted ten billion years ago, the redshift accumulated over its long journey is itself the record of ten billion years of vacuum light-speed evolution.

For nearly a century, the inherent limitation of our observational means has blinded us: ancient photons and their redshift are observable, but we possess ***zero*** direct evidence for the actual speed of light in the early universe. We can now, under the brand new LEM-SCDF paradigm, declare to the world:

Cosmic spatial expansion is fundamentally light-speed expansion. Redshift is merely the observable signature of global growing-$c(t)$ variation; the two are two facets of a single unified physical process.

Four centuries is infinitesimal against the cosmic evolutionary timescale. Since Galileo (1638) and Rømer (1676) first pointed out that light propagates at finite speed, generations of measurements have continually reinforced our conviction: the speed of light is a universal constant, set by the Creator, unchanging!

We are now beginning to realize that, to the cosmic giant, four hundred years is merely a single heartbeat. If the speed of light is not decreed by a Creator, but instead shaped collectively by every milligram of dust and every joule of energy in the universe — this is, in fact, a more harmonious and beautiful cosmos.

The Universe as a Coherent Physical Whole In conventional physics, no intrinsic mechanism universally connects all cosmic mass and energy, so the universe remains merely a vast collection of isolated local grains of sand, lacking global coordination. Yet each grain silently upholds its two sacred parameters—c and G—unchanging, as if decreed by the Creator.

The LEM-SCDF field provides a physical mechanism that instantaneously links every particle and photon through synchronized modulation. It encodes the distribution of all coupled mass-energy and delivers large-scale cosmic variations uniformly to every participating component. The universe thus becomes a dynamically coherent physical whole, unified by real, continuous field interaction.

Previous definitions of the universe were vague and imprecise. With the introduction of LEM-SCDF, the concept becomes strict and unambiguous: every elementary unit of mass and energy participates in shaping the evolution of the LEM-SCDF; the complete collection of all such mass and energy belongs to our defined universe, regardless of where it is distributed or how far it lies from any reference center.

Within this framework, cosmic expansion acquires a strictly well-defined physical meaning. Previously, a well-known logical paradox persisted: if the universe is already spatially infinite, what does Friedmann expansion describe? Expansion of an infinite scale? Within our new picture, the answer is clear: expansion describes the *statistical* enlargement of the distribution domain and the corresponding *statistical* dilution of the density inside the LEM-SCDF coupled system.

In the new picture, every photon experiences an active and continuous global modulation by the only entity that connects all cosmic constituents. This is not a decorative addition to the theory; it is the causal closure that the redshift phenomenon itself demands.

11. Discussion on Experimental Detectability of Tiny Evolving $c(t)$

Since the 1983 SI revision, the vacuum speed of light has been fixed to $c = 299{,}792{,}458\,\mathrm{m/s}$, rendering c no longer an independently measurable physical quantity. The constancy of c thereby serves as a fundamental axiom for the design and operation of nearly all metrological instruments.

Under the deductions of our growing-$c(t)$ model, the annual relative variation rate of $c(t)$ lies on the order of 10^{-11}. Such a tiny deviation surpasses the precision limit of all existing human-made measuring instruments and experimental techniques by at least two to three orders of magnitude.

Only leading experimental physicists, innovative measurement schemes and state-of-the-art facilities can potentially realize quantitative detection of the tiny evolving $c(t)$. The brief discussion presented in this section does not constitute any practically implementable experimental proposal.

Historical Measurements of the Speed of Light Historic interferometric campaigns documented in the 1987 NBS report [1] yielded a relative measurement ceiling of 3.67×10^{-9}. A series of historical experiments gradually improved the measurement precision of c via distinct physical mechanisms, summarized concisely below:

- Fizeau rotating toothed wheel (1849): Ground-based direct time-of-flight measurement of light, precision $\sim 10^{-4}$, the first quantitative terrestrial determination of c.
- Foucault–Michelson rotating mirror scheme (Foucault 1850, Michelson octagonal mirror refinement 1926): Optimized optical layout to suppress mechanical systematic errors, lifting measurement precision to $\sim 10^{-6}$.
- Microwave resonant cavity (1946–1970s): Calculated c from the identity $c = \lambda f$ by calibrating resonant frequency and cavity geometry; precision $\sim 10^{-9}$, the primary metrological standard prior to the 1983 SI redefinition.
- Stabilized laser interferometry (1972 NBS experiment): Separately measured the frequency and vacuum wavelength of frequency-stabilized lasers via atomic clocks and interferometric techniques, combining results to compute $c = \lambda\nu$; the experiment achieved a frequency measurement uncertainty of 6×10^{-10}, marking a 100-fold improvement over previous c determinations and laying the experimental foundation for the fixed definition of $c = 299\,792\,458$ m/s adopted by the 17th CGPM in 1983.

Estimated Variation Rate of the Growing-$c(t)$ Model Within the compact model, the derivative of $C(t)$ is given by Eq. (5.14). Taking the present moment as $t = 0$ and substituting the parameters k and β from Eq. (5.20), we obtain $\dot{C}(t) = 0.03145$, corresponding to $3.145 \times 10^{-11}\,\mathrm{yr}^{-1}$. Scaled to c_0, this yields a secular variation of $9.43\,\mathrm{mm{\cdot}s^{-1}{\cdot}yr^{-1}}$.

For the ultra-fine model, we utilize the `Planck18` standard redshift dataset and carry out calculations with z_1 from Eq. (6.1), taking the average value over the latest 10^4 years. This configuration yields a relative growth rate of 6.89% per Gyr, or equivalently $6.89 \times 10^{-11}\ \mathrm{yr}^{-1}$. When normalized to c_0, this corresponds to an annual variation of $20.7\ \mathrm{mm{\cdot}s^{-1}{\cdot}yr^{-1}}$.

When evaluating the rate on daily timescales, the above fractional yearly variation rates are divided by 365, yielding fractional daily growth rates spanning the interval $[0.86, 1.89] \times 10^{-13}\ \mathrm{day}^{-1}$, which translates to a daily increment in c of $[+26, +57]\,\mu$m/s.

Note: *Beyond the foregoing precision constraints, another critical concern lies in avoiding circular reasoning. Since the metrological standard definition* `Metra` *and photon wavelength* λ

are built upon the fundamental axiom of invariant c, entirely new measurement frameworks must be devised.

HEPS Device Conditions for Exploring Tiny $c(t)$ Evolution The High Energy Photon Source (HEPS) in Huairou, Beijing, is a cutting-edge fourth-generation synchrotron radiation facility that provides high-precision, high-resolution experimental conditions [17]. The total construction cost of HEPS is publicly reported to be approximately $700 million USD. Its key performance advantages are summarized below:

- HEPS hosts a 6 GeV high-energy electron storage ring, generating high-purity hard X-ray beams for stable calibration of physical parameters.
- It supports atomic-scale charge density mapping, single-nanoparticle imaging and high-coherence diffraction, breaking the limits of traditional fringe-counting measurement methods.
- Its fully oil-free ultra-high-vacuum system maintains a base pressure below 1×10^{-9} Pa. Such a setup removes refractive interference from air and supplies pure vacuum environments for light-speed-related experiments.
- The facility's nanoprobe achieves a focused spot size of 9.5–12.5 nm, providing atomic-level spatial resolution for precise calibration of optical paths and lattice geometries.
- The beamline delivers picosecond temporal resolution (10^{-12} s), upgradable to 100 fs with ultrafast detectors for high-precision timing of faint transient physical signals.
- HEPS uses atomically polished Kirkpatrick–Baez (KB) folding focusing mirrors with sub-angstrom surface flatness (< 0.1 nm), minimizing optical path distortion.

Whether the experimental community at HEPS will adopt the growing-$c(t)$ conjecture and further design dedicated experimental schemes remains undetermined.

Mission-M Research Initiative Under the prevailing physical paradigm with the constant speed-of-light assumption, it would be practically infeasible to obtain national public research funding dedicated to experiments searching for time-dependent variation of $c(t)$. The author therefore intends to lead and launch a private, non-governmental research initiative named *Mission-M*, with a budget of $75 million USD and a planned duration of thirty years. The mission is organized around three core principles: to *Measure* the speed of light with unprecedented precision, to *Model* the underlying physics should any variation be detected, and to *Mystery* the unknown regardless of the final outcome.

The first decade is devoted to theoretical analysis, experimental scheme design, and the invention and development of novel instruments. The remaining two decades would be dedicated to continuous observational campaigns. The overarching goal is to push the measurement precision of the speed of light to the 10^{-14} to 10^{-15} level, which represents an enhancement of 10^5 to 10^6 times relative to the 1983 SI reference, with a high-caliber team of experimental physicists to be assembled in China.

Two outcomes are conceivable. Even if the measurement confirms that the speed of light remains constant at this enhanced precision, the experiment would still constitute a valuable scientific result, analogous to the landmark 1887 Michelson–Morley experiment, which ruled out the existence of static ether and ether wind. By contrast, should a daily fractional variation near 10^{-13} be detected, it would open extensive new avenues for fundamental-physics investigations.

12. A Fresh Perspective on Outstanding Problems via the Growing-$c(t)$ Framework

In this section, we will re-examine several outstanding problems under this growing-$c(t)$ model, including the DESI DR2 dynamical dark energy preference, the cosmic photon budget crisis, the early galaxy stellar mass tension, the Population III star timescale conflict, the early solar lithium abundance puzzle, the late emergence of advanced intelligent life, and several other open issues.

Alternative Explanation for the DESI DR2 Tension In the standard ΛCDM paradigm, dark energy acts as a homogeneous background fluid driving cosmic acceleration [18]. However, recent DESI DR2 BAO measurements, combined with type Ia supernovae and CMB data, reveal a persistent deviation from a pure cosmological constant [19].This deviation is not an isolated finding. As recently highlighted by Abreu and Turner [20], the DESI data imply a maximum in the dark energy density around $z \simeq 0.5$, a behavior difficult to reconcile with a simple cosmological constant and hinting at new physics. The dark energy equation of state $w(z)$ is found to vary with redshift: at high redshift ($z \sim 0.75$), the data favor $w < -1$ (the phantom regime); at low redshift ($z \lesssim 0.2$), the data favor $w > -1$ (the quintessence regime). This crossing of $w = -1$ corresponds to the "quintom B" behavior, with a significance exceeding 3σ and a Bayesian evidence of $\Delta \ln E = 5.2 \pm 0.7$ in favor of dynamical dark energy over ΛCDM.

Within our growing-$c(t)$ framework, this observed pattern admits a straightforward physical interpretation. The revised Einstein field equation in Section 8 reveals that cosmic expansion couples directly to the growth of light speed $c(t)$. As matter aggregates into dense clumps, intergalactic volumes become tenuous, reshaping the global electromagnetic background via the LEM-SCDF field. According to $G(t) \propto c(t)^{-2}$, the steady growth of $c(t)$ weakens the effective long-range gravitational coupling. Crucially, the crossing behavior $w(z) < -1$ at high redshift and $w(z) > -1$ at low redshift is not exotic physics. It is the natural consequence of the monotonic decrease of $G(t)$ as $c(t)$ grows, which alters the expansion history and, when interpreted through the standard w_0-w_a parameterization, yields precisely the phantom-crossing pattern seen in DESI DR2. In this picture, "dynamical dark energy" is not a mysterious fluid, but a misidentification — the macroscopic signature of a universe whose gravitational coupling and light speed evolve together, locked by the invariant $P = G(t)c(t)^2$.

▷▷▷ ***Paradox:*** *Does dark energy fear humanity and actively evade the Solar System?* I have carefully examined the recent DESI DR2 observational data and uncovered a fundamental paradox hidden within standard cosmology, a paradox that to our knowledge has never been rigorously confronted in the literature. This paradox cuts to the heart of modern cosmology. Stated plainly:

- Stellar systems with identical evolutionary ages, approximately $t^\star = -4.67$ Gyr, exhibit utterly distinct dark energy behaviors depending solely on their spatial separation from our observational vantage point;
- Locally, within our 4.67-billion-year-old Solar System, dark energy signatures are practically non-existent;
- In distant cosmic environments, however, structurally analogous and coeval systems, also with $t^\star \approx -4.67$ Gyr, display pronounced dark energy effects, and the effect grows stronger with increasing distance;
- No existing ΛCDM mechanism can physically justify this unnatural cosmic spatial bias;
- Same cosmic age, drastically different physics. The only variable is distance from Earth and Solar System.

We therefore pose the question anew: does dark energy truly exist as a fundamental component of the universe? Or is its apparent behavior merely a mirror of our own observational position and light travel time?

The invariant relation $G(t)c(t)^2 = P$ may resolve this inconsistency. The apparent variation attributed to dark energy could naturally arise as an observational artifact of the cosmic evolution of $c(t)$ and $G(t)$. We do not claim to rule out the cosmological constant entirely; rather, we point out that an independent physical route to cosmic acceleration exists within the growing-$c(t)$ framework. The DESI DR2 observations, which challenge the canonical ΛCDM model, are fully consistent with this evolutionary picture and may represent the first observational evidence supporting the $G(t) \propto c(t)^{-2}$ scaling law proposed in this work. ◁◁◁

Note: *We apply the same cautious stance to the dark matter framework. Stronger effective gravity from diminished early $c(t)$ can partially explain high-redshift gravitational shortfalls. However, our mechanism cannot resolve the 400-fold mass mismatch observed in the low-redshift Coma Galaxy Cluster [21].*

Reinterpreting the Cosmic Photon Budget Crisis at $1 + z = 25$ Standard ΛCDM rests on two premises: a hot Big Bang cooling scenario, and the tSZ thermal scaling relation $T(z) = T_0(1 + z)$ with $T_0 = 2.725$ K. This relation predicts an intergalactic temperature of 68 K at $1 + z = 25$ and 19 K at $1 + z = 7$, where the former temperature is approximately 3.58 times the latter.

Per atomic physics, photons above 13.6 eV ionize hydrogen instantly; recombination speeds up sharply at lower gas temperatures. At $1 + z = 25$, early galaxies emit massive far-ultraviolet photons matching the 91.2 nm ionization threshold. Standard theory uses the 68 K temperature to argue recombination is slow, and abundant UV photons would fully ionize intergalactic hydrogen at high redshift.

This prediction clashes with joint observations [22]. JWST detects extreme UV photon output at $1+z=25$, while 21 CM radio signals prove widespread neutral hydrogen persists at the same redshift. This inconsistency forms the photon budget crisis, which standard cosmology only resolves near $1+z=7$ via falling UV production and faster recombination.

The severity of this crisis is underscored by recent efforts to resolve it within the standard paradigm. For instance, Austin et al. [23] and other recent studies [24] have attempted to reconcile JWST observations with reionization by invoking strong feedback mechanisms or HII clumping. While these attempts provide valuable insights, they often require fine-tuned parameters. The growing-$c(t)$ framework, by contrast, offers a more fundamental resolution by decoupling $T_{\rm rad}$ and $T_{\rm gas}$, as we have argued.

The $T(z)=T_0(1+z)$ formula assumes fixed light speed, conflicting with our growing-$c(t)$ model where $c(t)=c_0/25$ at $1+z=25$ and $c(t)=c_0/7$ at $1+z=7$. We separate two unrelated temperature quantities:

- $T_{\rm rad}(z)$: Effective blackbody temperature of distant photons measured by the tSZ effect;
- $T_{\rm gas}(z)$: True thermodynamic temperature of diffuse intergalactic hydrogen controlling ionization and recombination.

Standard cosmology incorrectly equates $T_{\rm rad}(z)$ and $T_{\rm gas}(z)$; our framework decouples them entirely.

Under our ZLZT cold primordial state, $T_{\rm gas}(z)<10$ K at both $1+z=25$ and $1+z=7$. The ultracold medium creates an extremely high recombination rate. Ionized particles rapidly recombine into neutral hydrogen after each photon collision, naturally reconciling all observational data without artificial fitting parameters.

Another possible reason is that the ΛCDM model makes incorrect estimates of the mass and density of stars at the $1+z=25$ epoch, leading to a serious overestimation of their UV photon output. This point will be analyzed in the next subsection.

Note: *This text focuses solely on the reionization-related narrow photon budget paradox, separate from the full-band EBL problem. We accept tSZ measurements but reject the unproven identification of radiative temperature and intergalactic gas temperature.*

Re-evaluating Stellar Mass and Density Within standard Newtonian celestial mechanics and conventional cosmology, all stellar and galactic mass calibrations fundamentally rest on the gravitational constant G_0, which is calibrated locally within the Solar System. Standard cosmological frameworks implicitly treat $G=G_0$ as invariant across all cosmic redshifts, and all subsequent calculations of stellar density, radius and stellar evolution obey this fixed scaling rule.

Recent JWST observations have identified candidate compact galaxies at extremely high redshifts ranging from $z\sim10$ to $z\sim25$. Under the assumption of a fixed gravitational constant G_0, the estimated average density of objects in this high-redshift regime approaches or even exceeds the nuclear saturation density [25].

If density exceeds this saturation threshold, runaway nuclear fusion ignites inside stars, driving their rapid collapse into black holes. This theoretical outcome conflicts with the stable,

long-lived high-redshift systems observed by space telescopes. Standard theories seek to resolve this tension by invoking unconfirmed physical mechanisms: super-Eddington accretion or the direct collapse of massive primordial gas clouds; however, neither scenario has been validated via local astronomical observations.

Turning to Newtonian celestial mechanics, orbital motions of planets, binary stars and virialized galaxies only constrain the combined product GM. This degeneracy arises directly from conservation of angular momentum. For a two-body gravitational system with reduced mass $\mu = \frac{Mm}{M+m}$, orbital angular momentum takes the form $L = \mu\sqrt{GMa(1-e^2)}$, where a and e represent the orbital semi-major axis and eccentricity, respectively.

Since only GM appears in the expression, orbital, angular-momentum and light-curve observations cannot separate G from the central mass M. For closed systems with conserved angular momentum, any (G, M) pair with fixed GM yields identical orbits, periods and velocity profiles.

Our growing-$c(t)$ model hypothesizes that the gravitational constant $G(t)$ rises with redshift following the relation $G(t) \propto c(t)^{-2}$. This indicates gravitational coupling was far stronger in the early high-redshift universe. If we retain the orbital observational invariant $G(t)M(t)$, the intrinsic baryonic mass $M(t)$ and density of celestial objects will be much lower than the predictions from standard astronomical models.

The cosmic scaling law of the growing-$c(t)$ framework breaks the degeneracy between G and M, shifting stellar density distributions to milder, physically consistent ranges compatible with the nuclear saturation limit of baryonic matter.

The growing-$c(t)$ model introduces a previously overlooked fundamental scaling degree of freedom into the standard ΛCDM analysis framework. Future work requires joint constraints combining atomic scaling derived from the Bohr radius, primordial nucleosynthesis, and binary orbital kinematics to resolve the GM degeneracy and establish revised mass scaling relations valid across all cosmic epochs.

Resolving Timescale Conflict for Population III Stars Within the standard ΛCDM model, the first generation of metal-free primordial stars—Population III stars—inevitably form with extremely large masses, ranging from ten to several hundred solar masses, due to the absence of metal-line cooling in primordial gas clouds. The standard ΛCDM model predicts that these very massive Population III stars remain on the main sequence for merely two to three million years [26].

This canonical prediction creates a severe timescale conflict with recent JWST observations of celestial objects in the high-redshift range $z \sim 10$ to $z \sim 25$. JWST has identified fully formed, massive galaxies that emerged merely several hundred million years after the Big Bang. If all these primordial massive stars burn through their nuclear fuel and die within just a few million years, galaxy assembly, supernova feedback, and metal enrichment must proceed at an unphysically rapid pace to produce the observed mature galaxies.

To alleviate this conflict with JWST observations, cosmologists have proposed unconfirmed ad-hoc physical hypotheses, including dark matter annihilation heating inside stellar cores and strongly top-heavy initial mass function (IMF) distributions.

We argue that the extremely short lifetimes of primordial stars in the standard ΛCDM model originate fundamentally from an untested assumption: that the gravitational constant $G = G_0$

remains invariant across all cosmic epochs. This fixed-G prior systematically overestimates the baryonic mass of high-redshift primordial objects.

Our growing-$c(t)$ framework follows the relation $G(t) \propto c(t)^{-2}$, indicating that gravitational coupling was significantly stronger at high redshifts in the early universe. Under the constraint of conserved orbital observables, the enhanced early $G(t)$ necessarily reduces the intrinsic baryonic mass $M(t)$ of primordial stars. Crucially, this mass reduction effect dominates over the strengthened gravitational coupling, yielding a substantially weakened core compression in early stars.

Stellar physics establishes a robust mass-luminosity relation $L \propto M^{3\sim4}$, implying that stellar energy consumption rises superlinearly with mass. The sensitivity of stellar evolution to the value of the gravitational constant has been examined in classical studies [27], which demonstrated that a change in G directly alters the nuclear burning timescale. A moderate reduction in intrinsic stellar mass — as implied by a larger $G(t)$ in the early universe — dramatically suppresses the nuclear burning rate, whereas the total nuclear fuel content decreases only linearly.

Therefore, the growing-$c(t)$ framework may resolve this severe timescale conflict without invoking exotic heating mechanisms or fine-tuned initial mass distributions. By breaking the GM degeneracy and revising the high-redshift stellar mass scaling, early stellar evolution timescales return to physically consistent ranges, offering an elegant and self-consistent explanation for the mature galactic structures observed in the extremely early universe.

The Early Solar Lithium Abundance Puzzle Standard stellar evolution theories assume a constant speed of light c. Based on this premise, they predict a definite primordial lithium abundance for the early Sun. However, high-precision spectroscopic observations have long revealed a marked discrepancy: the lithium content on the solar surface is merely one-third of the theoretical prediction. This well-known inconsistency is referred to as the long-standing solar lithium puzzle.

To fix this problem, existing theories add extra assumptions. These include stronger internal mixing, early mass loss, or lithium hiding in planets [28]. However, there is no direct proof for these ideas, and they fail to provide a unified, self-consistent physical picture.

Within the growing-$c(t)$ framework proposed in this study, the Solar System formed at a cosmic redshift of $1+z \approx 1.43$. Following the scaling law $c(z) = c_0/(1+z)$, the vacuum light speed at the epoch of solar formation was roughly 70% of its present-day value. Correspondingly, the Bohr radii of hydrogen, lithium, carbon, and all other elements shrink to 70% of their present-day sizes, imposing inherent limits on the initial lithium reservoir of the protosun.

Lithium is a light element. Lower $c(t)$ and smaller Bohr radii together make it easier for lithium to start nuclear burning. Lithium atoms quickly burn away in the young Sun, so much lithium is lost early on. This process creates a solar surface lithium level much lower than calculations that use a fixed speed of light.

Many young stars lie within ten light-years of the Sun. These stars formed at redshifts near $z = 0$, when the cosmic speed of light was nearly equal to today's value. Our model gives a testable prediction: nearby young stars lose far less lithium, so their surface lithium levels will be clearly higher than the Sun's.

Earth's Biological Evolution and the $G(t)$ The Cambrian Explosion and the Carboniferous carbon burial represent two landmark episodes in the history of Earth's biological

evolution. In the standard geological and paleontological frameworks, these events are primarily interpreted through planetary-scale internal processes—such as supercontinent breakup, continental weathering rates, oceanic redox stratification, the evolutionary timing of lignin-degrading fungi, and swamp sediment preservation conditions.

According to the prevailing view, for nearly three billion years prior to the Cambrian, Earth hosted only simple life forms, with evolutionary progress largely suppressed by multiple environmental thresholds. It was not until the narrow window of 25 million years (540–515 Ma) that the critical planetary conditions simultaneously crossed their evolutionary thresholds, triggering the Cambrian radiation, during which most major animal phyla emerged in a geologically brief interval.

The Carboniferous period featured a uniquely stable planetary climate that persisted for as long as 60 million years (360–300 Ma). The warm, oxygen-rich environment facilitated large-scale geological carbon sequestration; the subsequent decline in atmospheric oxygen levels eliminated giant organisms with limited brain capacity, thereby opening ecological niches for the evolution of mammalian brains.

The evolutionary record of Earth thus exhibits a striking asymmetry: billions of years of prolonged stagnation punctuated by abrupt biological transitions within extremely short geological windows. Many current theories attribute the emergence of intelligent life to random coincidences—the so-called *lucky Earth* hypothesis.

Against the background of continuously evolving $G(t)$ and $c(t)$, we tentatively suggest that a comprehensive picture may need to incorporate nested constraints across three scales: *cosmic, stellar, and planetary*. In particular, the time-varying gravitational coefficient $G(t)$ may have undergone key relaxation processes during the Cambrian and Carboniferous epochs.

As gravitational coupling approached stability, orbital perturbations within the Solar System were minimized, fluctuations in surface solar flux were substantially damped, and Earth's rotation and precession cycles entered a prolonged phase of dynamical regularity.

While planetary-scale analyses remain essential, incorporating the macroscopic boundaries set by the cosmic background fields $G(t)$ and $c(t)$ may offer a broader context.

One might then be more inclined to view the appearance of human intelligence on Earth at this particular moment not as a pure accident, but perhaps as a contingent outcome made possible by a sequence of necessary physical conditions. This interpretation, however, remains highly speculative, and we present it with due caution.

LLR Does Not Prove an Exact Constant G The most precise direct laboratory measurement of G was reported by the Luo-Jun group (2018) [29], achieving a relative uncertainty of 11.6×10^{-6} using two independent Cavendish-type torsion-balance schemes.

By contrast, the constraint $\Delta G/G \sim 10^{-13}\,\mathrm{yr}^{-1}$ from Lunar Laser Ranging (LLR) is not a direct measurement, but an inference built upon the prior model assumption that light speed c is constant [30]. We now dissect the reasoning behind this inference.

After removing known perturbative effects from the Sun and other planets, LLR observations yield an apparent lunar recession rate of $\Delta D = \mathbf{3.8}\ \mathrm{cm\,yr}^{-1}$ under the constant-c assumption [31]; the standard LLR model attributes this entire recession to tidal angular-momentum

dissipation within the Earth-Moon system. This corresponds to a fractional growth of the round-trip light travel time:

$$\frac{\Delta T}{T} = \frac{\Delta D}{D} = \frac{3.8 \text{ cm/yr}}{3.844 \times 10^{10} \text{ cm}} = 9.89 \times 10^{-11} \text{ yr}^{-1} . \tag{12.1}$$

Note: *Crucially, ΔD is not directly measured; LLR measures only the round-trip light travel time $2T$ and its annual variation $2\Delta T$. The recession distance ΔD is then inferred from ΔT by assuming a constant speed of light c.*

As stated in Section 11, within our ultra-fine model, $\Delta c/c = 6.89 \times 10^{-11}$ yr^{-1}. Since $T = D/c$ (single-trip light travel time), logarithmic differentiation gives the real orbital expansion ratio:

$$\frac{\Delta D_{\text{real}}}{D} \approx \frac{\Delta T}{T} + \frac{\Delta c}{c} , \tag{12.2}$$

$$\frac{\Delta D_{\text{real}}}{D} \approx 9.89 \times 10^{-11} + 6.89 \times 10^{-11} = 16.78 \times 10^{-11} \text{ yr}^{-1} , \tag{12.3}$$

which corresponds to a physical recession rate:

$$\Delta D_{\text{real}} = 3.8 \times \left(1 + \frac{6.89}{9.89}\right) \approx \mathbf{6.45} \text{ cm yr}^{-1} . \tag{12.4}$$

Thus, under our ultra-fine growing-$c(t)$ model, the Moon recedes from Earth at **6.45** cm yr^{-1} — significantly larger than the 3.8 cm yr^{-1} inferred from standard LLR analysis.

Furthermore, based on the core relation $Gc^2 =$ constant adopted in this work, first-order differentiation yields the time-variation relation for the gravitational constant:

$$\frac{\Delta G}{G} = -2 \frac{\Delta c}{c} = -13.78 \times 10^{-11} \text{ yr}^{-1}. \tag{12.5}$$

Under the condition of angular-momentum conservation for the Earth-Moon two-body system, $GR =$ constant, which gives

$$\frac{\Delta D_{\text{gravity}}}{D} = -\frac{\Delta G}{G} = 13.78 \times 10^{-11} \text{ yr}^{-1} . \tag{12.6}$$

Note: *This component of orbital expansion arises purely from the cosmic evolution of gravity G, independent of tidal dissipation.*

Subtracting the cosmic contribution from the total rate, the remaining part corresponds to the orbital expansion induced by ***tidal*** dissipation. So, we have:

$$\frac{\Delta D_{\text{tide}}}{D} = \frac{\Delta D_{\text{real}}}{D} - \frac{\Delta D_{\text{gravity}}}{D} = \mathbf{3.00} \times 10^{-11} \text{ yr}^{-1} . \tag{12.7}$$

$$\Delta D_{\text{tide}} = 6.45 \times \left(\frac{3.00}{16.78}\right) \approx \mathbf{1.15} \text{ cm yr}^{-1}. \tag{12.8}$$

$$\Delta D_{\text{gravity}} \;=\; 6.45 \times \left(\frac{13.78}{16.78}\right) \;\approx\; \mathbf{5.30}\ \text{cm}\,\text{yr}^{-1}. \tag{12.9}$$

Thus, in contrast to the standard LLR model, which attributes the entire observed recession to tidal dissipation, our variable-c model offers a distinctly different interpretation. In our model, ΔD_{tide} is only 1.15 cm yr^{-1}, significantly smaller than the 3.8 cm yr^{-1} inferred from the standard LLR framework.

Consequently, we conclude that the 10^{-13} bound on $\Delta G/G$ from LLR model cannot be considered a solid constraint against the conjecture presented in this work.

The Casimir Effect Is Not Evidence for a Vacuum Energy Catastrophe In conventional quantum field theory, the vacuum is assumed to contain a substantial quantum zero-point energy, and the Casimir force is interpreted as definitive experimental proof of its existence. The well-measured scaling relation $F \propto D^{-4}$ at nanometer-scale separations is taken as direct evidence that such vacuum fluctuations fill all of empty space.

If this vacuum energy were indeed present everywhere, its density would exceed the observed cosmic dark energy density by a factor of order 10^{120}. This is the so-called *vacuum energy catastrophe*, long treated as an unsolved fundamental crisis within the standard ΛCDM framework.

However, this catastrophe is not a direct observational result. Jaffe (2005) [32] demonstrated that Casimir forces may be derived without invoking any global zero-point vacuum energy; the effect only measures an energy difference between two boundaries, not an absolute vacuum energy.

The present work follows and extends Jaffe's argument. We reinterpret the Casimir effect within the LEM-SCDF background field framework and carry the reasoning further into cosmological dark-energy considerations.

The severe theoretical dilemma originates entirely from an unreasonable and unphysical extrapolation. Localized boundary effects measured between two material plates are incorrectly extended by existing mainstream theories into a uniform vacuum property filling the entire universe. In the LEM-SCDF background field picture proposed here, the Casimir force arises from localized compression of the universal background field induced by adjacent material boundaries. Compression between the plates necessarily triggers complementary rarefaction in the surrounding space, establishing a globally conservative redistribution of the background field density. The observed force merely reflects this local imbalance. The force manifests only under restrictive laboratory geometry, and no empirical evidence confirms that such extreme vacuum energy uniformly fills cosmological empty space.

Once this artificial global extrapolation is abandoned, the vacuum energy catastrophe disappears. The Casimir effect can no longer serve as valid evidence for dark energy, and there is no longer any need to introduce hypothetical dark-energy components to rescue cosmic observations.

Local Renewal Misread as Cosmic Creation Standard cosmology borrows what we see today to explain what happened at the beginning. Supernovae and black holes are local

events in the mature universe, yet researchers extrapolate them backward in time to describe the early cosmos.

This logical mismatch between global early-universe phase transitions and local late-universe events has been noted before: Milne's cosmological principle [33] emphasizes the global uniformity of the early universe, while Barrow and Tipler [34] underscore the fundamental distinction between the extreme physical conditions of the early epoch and the late-time local environment.

This constitutes a fundamental logical mismatch. All observed supernovae and black holes originate from the late universe, after the formation of atoms, galaxies, and stars. In contrast, the early universe evolved through hadronization, plasma coupling, and CMB decoupling stages, with no stars, no compact objects, and no gravitational collapse.

Early universe evolution is global: the entire cosmos undergoes phase transitions as one unified system. Hadronic substance, then plasma, then neutral atoms and the CMB — this sequence built the fundamental material foundation of the cosmos. It is the true creation of all baryonic matter.

Black holes and supernovae are purely local. They occur only after the cosmic background is fully in place. They redistribute mass, release energy, and recycle stellar material. They are renewal processes within galaxies — not creation processes of the universe itself.

We should draw a clear line between these two distinct physical levels. This distinction will remove a long-standing conflation, and provides a physically consistent basis for understanding early cosmic structure formation.

On Reconstructing the Friedmann Equations The standard Friedmann equations, Eq. (1.14) and Eq.(1.15), were derived from the Einstein field equations. With the revised $G_{\mu\nu}$ now given by Eq. (8.2), we need to reconstruct Friedmann equations and apply the necessary corrections to the `Planck18` redshift data.

Methodological Reflection on Modern Cosmology It has been argued by some commentators that modern standard cosmological frameworks risk degenerating into what may be termed "*Sapeur physics*": elaborate mathematical formalism serves as ornate outer garments, while the foundational physical premises rest on unconfirmed, stacked hypotheses. When observational tensions emerge within the ΛCDM paradigm, the conventional remedy adopted is to introduce extra free parameters and ad-hoc components to reconcile observational mismatches. This iterative remedial process is analogized to *overfitting*, whereby countless corrective patches are piled onto an inherently flawed underlying theoretical structure.

We should revisit the foundational principle highlighted by Heisenberg and restated by Yang Chen-Ning: physical theories ought to be built around observable quantities. Intermediate theoretical constructs do not need to be directly measurable. For example, spacetime curvature in general relativity cannot be detected via a single standalone instrument, yet its derived outcomes—light deflection, gravitational redshift and gravitational waves—are quantitatively verifiable observables. This line of reasoning maintains a solid empirical foundation.

A representative benchmark for this rigorous standard is the Copenhagen interpretation of quantum mechanics. Though its intuitive physical picture strikes many as counterintuitive

and hard to visualize, the framework delivers fully tractable mathematical calculations and precise, testable physical predictions across all quantum experimental scenarios. Its abstract internal logic never detaches from measurable spectral, spin and interference observables, which distinguishes it from purely speculative multi-layered hypotheses in contemporary cosmology.

Based on this criterion, a critical boundary must be drawn. One may tolerate a single layer of unobservable intermediate entities bridging theory and measurement. It is philosophically unsustainable to nest successive tiers of untestable conjectures, where every new layer lacks independent observational anchoring. Such an approach abandons the empirical core of physics. A theoretical model cannot rely indefinitely upon unobserved hypothetical ingredients merely to accommodate existing data.

The present work also strictly constrains itself to this empirical discipline. Although the extremely subtle time-varying behavior of the speed of light $c(t)$, with variation magnitudes approximately at the 10^{-13} to 10^{-14} level, cannot be directly measured in the near future within the next decade, the proposed framework still yields concrete, falsifiable predictions for multiple cosmic observables, including cosmic redshift evolution, the epoch-dependent probability of black hole formation evolving alongside $c(t)$, and the redshift-dependent dark energy equation of state $w(z)$. All theoretical inferences in this study ultimately trace back to astronomically accessible observational quantities.

Postscript

A Hymn to the Breathing Cosmos and Intelligent Life In its earliest days, the cosmos and our blue Earth were clumsy novices. Massive ancient stars burned brightly and collapsed suddenly, too short-lived to nurture planetary systems. Our planet once had a frozen equator and endured five devastating mass extinctions. Life rose, fell, and reset again and again amid countless trials. Every collapse is not an end, but a new beginning. Every setback gathers silent power for the final bloom of life.

After billions of years of evolution, fundamental physical quantities of the universe fell into perfect balance. The speed of light, gravity, Earth's 23.5-degree axial tilt, atmospheric oxygen and the ozone layer all settled at ideal states to sustain life. This is the true meaning of the breathing cosmos: it runs on the endless rhythm of time. Stars, planets and all living things breathe with the universe, and intelligent life is born of this eternal rhythm.

May the light of intelligence never fade. Earth has waited four billion years, and the universe far longer, for conscious life to emerge. May human civilization live up to this ancient waiting, turning a fleeting spark into lasting radiance across the galaxy. We are fragile yet infinitely precious. We are not temporary travelers in the universe. Instead, we are the *very eyes* through which the cosmos, after billions of years of silence, finally sees and understands itself.

On the Origin of This Work I first began wondering about the essence of light and gravity decades ago. A sudden insight came to me during a drive in early March. Realizing this is one of the deepest mysteries of physics, I dared not even entertain the thought. In early

May, the growing passion urged me to write this study. I attempt to build a logically consistent physical framework in this work. Even if my theory fails to match future observational evidence, I hope it can serve as an imaginative thought experiment.

Appendix: What if the Conjecture Holds True

Disclaimer: Addressing many long-standing puzzles is merely an entry ticket. It does not constitute proof that the framework is physically correct. This appendix presents only conceptual thought-experiment reasoning.

If its core postulates hold true, this framework would fundamentally challenge the ΛCDM standard cosmological paradigm. Its impacts would extend across general relativity, particle physics, astrophysics, metrology, and scientific methodology. It may resolve twenty to thirty long-standing observational and theoretical conflicts, as summarized below.

This framework builds upon only two concise core postulates:

- Macroscopic Postulate: An invariant cosmic coupling constant $\boldsymbol{P}$ exists. The relation $G(t)c(t)^2 = \boldsymbol{P}$ holds over the whole cosmic history.
- Microscopic Postulate: The fine-structure constant $\boldsymbol{\alpha}$ remains invariant across all cosmic epochs.

Standard ΛCDM invokes dark energy, dark matter, and inflation with layered auxiliary assumptions to accommodate observational tensions. Instead of introducing multiple additional free parameters, the growing-$c(t)$ model uses a single evolving variable $c(t)$ to address numerous open physical puzzles. This is its main conceptual strength.

I. Cosmology: Ten core puzzles discussed in the manuscript

- **01**_ *Causal mechanism behind cosmological redshift.* Standard cosmology offers a geometric picture: space expands and stretches photon wavelengths. Yet it cannot answer the causal question: after photons decouple from emitters and travel through near empty space, what physical agent continuously informs photons of cosmic expansion? Within this model, cosmological redshift is the observational signature produced by cosmic evolution of vacuum light speed. The globally synchronized LEM-SCDF background field provides this missing causal link.
- **02**_ *Rejecting the hot Big Bang initial condition.* The standard model assumes an initial state of extreme temperature and density. This conjecture adopts the ZLZT (Zero-Light-Speed and Zero-Temperature) primordial state. The universe evolves gradually starting from near-zero light speed, rather than from a violent explosion. The cosmic timescale before CMB decoupling becomes substantially longer than the 37 kyr predicted by ΛCDM.

- **03**_ *Ionization temperature paradox at CMB decoupling.* Standard cosmology predicts extremely high temperatures around the CMB epoch. This framework, by contrast, starts from a ZLZT primordial assumption, which leads to a light speed of approximately 272 km/s at the CMB epoch. The corresponding ionization temperature at recombination is only about 82.64 K, fundamentally revising the physical picture of cosmic recombination and reionization.

- **04**_ *Evolution of Schwarzschild radius and scarcity of planetary systems in high-redshift galaxies.* Since $r_s(t) \propto c(t)^{-4}$, Schwarzschild radii become enormously large in the early universe. Gravitational capture-cross-sections are strongly amplified. Early stars rapidly accrete surrounding material and erase protoplanetary disks. Planetary systems barely obtain chances to form. Only when $c(t)$ rises and $r_s(t)$ shrinks does the cosmic environment stabilize and allow widespread planet formation. This naturally matches JWST observations: high-redshift galaxies are star-dominated with very few planetary candidates. A heuristic formula Eq. (7.1) further delivers redshift-dependent predictions for planet-formation probability, testable with future observations.

- **05**_ *Timescale crisis of Population III stars.* JWST observes fully assembled massive galaxies at very high redshifts. Within standard ΛCDM, metal-free Population III stars possess extremely short main sequence lifetimes, and the fixed-light-speed timeline leaves insufficient time for galaxy assembly. In this model, however, early $G(t)$ takes larger values under the GM degeneracy: the inferred M is consequently smaller than standard estimates, so nuclear burning proceeds more slowly (though still relatively fast), and the effective physical timescale is stretched. What standard chronology counts as ~ 1 Myr may correspond to ~ 10 Myr or more near $1+z \sim 10$. True baryonic masses of primordial stars decrease, while nuclear burning timescales extend. No exotic stellar-core heating or extreme initial mass function assumptions are needed.

- **06**_ *DESI DR2 dynamical dark energy tension (quintom crossing behavior).* Standard cosmology struggles to interpret observed $w(z)$ crossing $w = -1$. This model invokes no real dark energy fluid. As $c(t)$ grows, $G(t)$ decreases monotonically and modifies cosmic expansion history. When analyzed with standard ΛCDM parameter pipelines, such evolution reproduces the apparent quintom-crossing signal. Apparent dynamical dark energy could turn out to be an observational artifact.

- **07**_ *Cosmic photon budget crisis during reionization.* JWST detects abundant far ultraviolet photons at $z = 25$, while 21 cm measurements show considerable neutral hydrogen coexisting at the same epoch. Standard models need finely tuned feedback and clumping factors for reconciliation. This framework strictly distinguishes radiative blackbody temperature $T_{\rm rad}(z)$ and gas thermodynamic temperature $T_{\rm gas}(z)$. Intergalactic gas stays extremely cold at high redshift and yields very high recombination rates. This relieves the tension without artificial fitting parameters.

- **08**_ *Solar lithium abundance puzzle.* The Solar System formed at $1 + z \approx 1.43$, when $c(t)$ reached roughly 70% of its present day value. Bohr radii shrank accordingly. Lithium underwent efficient early nuclear burning. This explains the observed deficit of solar surface lithium. One falsifiable prediction follows: nearby coeval young stars should show considerably higher surface lithium abundances.

- **09**_ *Constraints from Lunar Laser Ranging (LLR) no longer bound constant G.* The well-known LLR upper bound on $\Delta G/G$ is derived under the axiom of constant vacuum light speed. If $c(t)$ itself undergoes secular drift, round-trip light travel time signals mix two contributions: genuine lunar orbital recession plus cosmic variation of $c(t)$. Standard LLR limits cannot serve as rigorous evidence for a time-invariant gravitational constant.

- **10**_ *Revisiting the Dirac large-number hypothesis.* Dirac conjectured in 1937–1938 that fundamental constants might evolve with time, but he did not provide a self-consistent framework. In this framework, $G(t) \propto c(t)^{-2}$ and $N_{\mathrm{Dirac},1} \propto c(t)^5$. The numerical proximity is merely a transient feature, not a universal law.

Note: *All items below represent conceptual inferences rather than proven conclusions. They are not fully expanded within the main manuscript due to page constraints.*

II. Extended Astrophysics Puzzles

- **11**_ *Systematic mass-estimation bias for high-redshift galaxies.* Astronomical observables constrain only the product GM, and cannot separate G from baryonic mass M. All high-redshift galaxy and black hole masses are presently evaluated adopting the local Solar System value G_0. If $G(t)$ were larger at early epochs, intrinsic baryonic masses of cosmic objects would be substantially lower than standard estimates. Several tensions related to "supermassive black holes already existing at extremely high redshift" may be partially alleviated.

- **12**_ *Redshift-dependent properties of quasars and AGNs.* Enlarged early Schwarzschild radii enhance gravitational capture-cross-sections and gas accretion efficiency. This provides a new perspective for understanding why high-redshift quasars are exceptionally luminous, without exotic black hole seed formation scenarios.

- **13**_ *Galaxy rotation curves and part of the dark matter problem.* Enhanced gravitational coupling $G(t)$ in the early universe means ordinary baryonic matter can generate stronger gravitational effects. Part of the galaxy dynamics "missing mass" signal may stem from time-evolving $G(t)$. Nevertheless, the manuscript explicitly states this mechanism cannot fully account for low-redshift galaxy cluster mass discrepancies, and therefore dark matter cannot be completely ruled out.

- **14**_ *Full revision of stellar evolution diagrams.* Stellar structure equations directly depend on G. $G(t)$ differs at different cosmic redshifts. Zero-age main sequence tracks, mass luminosity relations, and pulsation behavior of variable stars all become redshift-dependent. Cepheid variables are a pillar of the cosmic distance ladder. If this conjecture holds, the whole distance scale must be recalibrated, offering a new route to resolve the Hubble tension.

- **15**_ *Gamma-ray bursts and supernova properties across cosmic time.* Type Ia supernovae are treated as standard candles. If $G(t)$ and atomic-scale quantities scaling with $c(t)$ evolve with redshift, supernova peak luminosity is not a universal constant. Conclusions derived from supernova-based cosmic acceleration analysis require re-evaluation.

- **16**_ *Big-Bang Nucleosynthesis (BBN) revisited.* Reaction cross sections and Coulomb barriers depend on $\varepsilon_0(t)$ and $G(t)$. Standard BBN calculations assume fixed c and G. Within this model, temperature-density conditions and cosmic timing for primordial nucleosynthesis change completely. New interpretive space opens for observed primordial helium and deuterium abundances.

- **17**_ *Casimir effect and the vacuum energy catastrophe.* Conventional quantum field theory treats the Casimir force as proof of a global zero-point vacuum energy, leading to the so-called vacuum energy catastrophe. The present work extends Jaffe's argument within the LEM-SCDF framework: the Casimir force arises from localized compression of the universal background field, necessarily accompanied by complementary rarefaction elsewhere, so that the total field density is globally conserved. Once this artificial global extrapolation is abandoned, the vacuum energy catastrophe disappears, and the Casimir effect can no longer serve as valid evidence for dark energy.

III. Fundamental Theoretical Physics

- **18**_ *Reinterpretation and revision of Einstein field equations.* The constant G inside the original field equations is replaced by $P/c(t)^2$. General relativity is no longer a purely geometric theory of gravity. Spacetime curvature dynamics couple with the vacuum electromagnetic background LEM-SCDF field. Mass-energy distribution, spacetime geometry, and electromagnetic vacuum parameters form a closed dynamical loop.

- **19**_ *Alternative pathway toward Unified-Field Theory (UFT), bypassing conventional gravity quantization dead-ends.* Traditional research programs try to quantize spacetime geometry directly and meet severe obstacles. This framework shifts dynamical degrees of freedom to the LEM-SCDF background field. One may prioritize quantizing this cosmic background field instead of quantizing gravity itself. It offers a modern realization of Einstein's 1920 Leiden lecture vision of a physical spacetime new-ether.

- **20**_ *Physical ontology for quantum-entanglement "spooky action at a distance".* The LEM-SCDF field propagates much faster than local light speed. Crucially, it transports no energy or causal signals; only global background state synchronization. It does not violate special relativistic local light-cone causality, yet provides a physical substrate for nonlocal quantum correlations. Entanglement is no longer merely a mathematical interpretive feature.

- **21**_ *Major upgrade over historic Variable-Speed-of-Light (VSL) models.* Earlier VSL proposals from Magueijo aimed to resolve horizon and flatness puzzles but worsened photon scattering pathologies. In contrast, the ZLZT primordial condition, with its near-zero initial light speed, circumvents those scattering problems from the very beginning.

- **22**_ *Revised foundational logic for the `SI` unit system.* The modern definition of the `Metre` is anchored to a fixed constant c. Under this conjecture, one must distinguish the present-day metrological metre L_m and the atomic metre L_a, defined by fixed atomic oscillation cycles whose physical spatial extent varies with $c(t)$. Many metrological measurements need reinterpretation.

- **23**_ *All Planck-scale quantities become derived evolving quantities.* Planck mass, Planck length, Planck density, and Planck force are no longer timeless universal constants. They obey power-law scalings with $c(t)$. The scale emerges as a property of the evolving cosmic background, not a fixed benchmark for quantum gravity physics. This reshapes starting assumptions for quantum gravity studies.

- **24**_ *Confinement Generates Substance: clear concept separation of substance versus energy.* Standard QCD interprets hadron mass as stored confinement energy. This work introduces M_{aps}, arche-particle-substance. Substance itself is conserved; $E = Mc(t)^2$ acts as a time-varying cosmic exchange rate between substance and energy. Substance and energy become conceptually distinct categories. This revises physical and philosophical understanding of mass-energy equivalence.

- **25**_ *The cosmos acts as one globally coherent entity.* Conventional physics treats the universe as an ensemble of disconnected local grains with externally imposed constants c and G. Within this picture, all mass and energy components collectively shape the LEM-SCDF background field. Every particle and photon maintains continuous global coupling. The universe becomes a dynamically coherent whole instead of a collection of isolated patches.

IV. Planetary-Science and Biological Evolution

- **26**_ *Redshift-dependent statistical predictions for exoplanet populations.* From the $r_s(z)$ scaling relation we obtain testable statistical expectations: planet occurrence fractions should be significantly suppressed at high redshift. As cosmic time proceeds, planetary systems gradually become more common. Future JWST and PLATO exoplanet surveys can statistically verify or falsify this trend.

- **27**_ *Evolution of primordial cosmic magnetic fields.* Since $\mu_0(t)$ and $\varepsilon_0(t)$ evolve together with $c(t)$, the whole history of cosmic magnetic field generation is rewritten. It provides new approaches for solving the origin of galaxy magnetic fields.

- **28**_ *Cosmic-background constraints upon Earth biological evolution.* As $G(t)$ slowly decays, Solar-System orbital perturbations and solar-flux variability decrease. Major evolutionary transitions such as the Cambrian explosion and Carboniferous events are not pure accidents of terrestrial geology. They may correspond to time windows where cosmic-scale gravitational background conditions relax to suitable ranges. The emergence of intelligent life is not purely random luck, but a contingent outcome enabled by gradual cosmic physical relaxation.

Conceptual Appeal Does Not Equal Physical Truth Addressing existing observational tensions is only a necessary condition, not sufficient proof. Explanatory power is an entry qualification, not a verdict. Decisive weight falls onto falsifiable predictions: the tiny daily fractional variation demands future facilities such as HEPS or the proposed Mission-M thirty-year program; statistical exoplanet occurrence versus redshift; observed properties of high-redshift stars and Cepheids; and higher precision DESI and future BAO measurements of $w(z)$.

Although the framework keeps its free parameter count low, it introduces two novel ontological postulates: the LEM-SCDF cosmic dark field and the M_{aps} arche-particle-substance. Direct

independent experimental evidence for these entities remains absent at present. A large body of follow-up theoretical work also remains open: the Friedmann equations, primordial nucleosynthesis, full stellar structure computations, and cosmic distance calibrations all require complete self consistent re-derivation. The present manuscript should be regarded as a program-forming outline; many derivations remain heuristic.

What if the Conjecture Holds True If this conjecture were confirmed, the ΛCDM standard model would become an effective approximate theory valid only for late cosmic epochs. The hot violent Big Bang would be replaced by the ZLZT picture of slowly growing light speed. Dark energy and portions of dark matter signals might turn out to be observational artifacts originating from evolving background constants. The familiar gravitational constant G would be demoted to an evolving derived quantity; the truly invariant cosmic quantities would be P and the fine structure constant α. Local Lorentz invariance of special relativity would be preserved, but on cosmic scales vacuum light speed would represent a time dependent global background. Revised Einstein field equations would link gravity and electromagnetism via the LEM-SCDF field, opening new directions for unified field theory research. Cosmic distance ladders, stellar masses, black-hole masses, supernova standard candle analyses, and primordial nucleosynthesis calculations would all require recalibration. Terabytes of archival JWST and DESI data would need re-analysis within the new paradigm. Even the foundational definitions of SI units, and the conceptual distinction between substance and energy, would be affected.

In short, this manuscript is not a minor patch to existing cosmology. It uses the single relation $G(t)c(t)^2 = P$ to address two to three dozen puzzles across cosmology, astrophysics, and fundamental theory. Nevertheless, in physics, elegant logical coherence serves only as a starting point. Final judgment belongs to telescopes and high-precision future experiments.

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v2.3  2026-09-14  Added Casimir effect and vacuum energy discussion to Section 12.
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v1.9  2026-08-18  Added LLR does not prove an exact constant G to Section 12.
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v1.7  2026-08-05  Added Einstein's Leiden lecture on vacuum materiality to Section 8.
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